

Design and Development of Charity System Model based on Blockchain Technology

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Abstract—Blockchain technology is emerging as a revolutionary solution for improving transparency, trust, and accountability in the charitable sector. This paper introduces a blockchain-based model for a charitable system aimed at resolving significant issues in conventional charitable frameworks, including insufficient transparency, elevated administrative expenses, and donor skepticism. The proposed model employs decentralized ledger technology and smart contracts to establish a secure, verifiable, and tamper-resistant donation process, enabling donors to monitor funds and guaranteeing that donations are allocated in accordance with specified conditions. This model enhances donor trust and mitigates operational inefficiencies by reducing intermediaries through the implementation of blockchain technology. The results demonstrate the model's capacity to enhance the integrity of charitable operations and underscore the significance of regulatory and accessibility factors for broad implementation.

Keywords—blockchain, smart-contracts, cryptocurrency, ethereum, decentralization, charity

I. INTRODUCTION

The charity sector is crucial in addressing social crises; nonetheless, it faces significant problems, including inadequate transparency, inefficiencies in fund administration, and donor distrust [1]. Traditional charity structures can demonstrate limited accountability, prompting concerns about the distribution and use of contributions. These issues undermine the effectiveness of philanthropic organizations and dissuade potential donors, especially as the scale of global charitable contributions increases. Blockchain technology, defined by its decentralized ledger, security, and transparency attributes, is emerging as a feasible answer to these difficulties. By employing blockchain technology, charitable organizations may establish transparent, unchangeable records of donations, thus enhancing operational efficiency and cultivating trust among stakeholders.

Blockchain technology is inherently decentralized, providing a digital ledger that records transactions across a network of nodes in an unalterable fashion. The decentralized framework ensures that data cannot be altered retroactively, therefore reducing the likelihood of fraud and improving traceability [4]. In a philanthropic setting, blockchain enables real-time tracking of gifts from donor to recipient, offering funders transparency into the allocation of their cash. This capacity is crucial for cultivating trust, as it allows donors to verify that their gifts are employed as intended. Furthermore, smart contracts—self-executing agreements with predefined conditions—can automate and enforce regulations for cash

allocation, ensuring that donations are disbursed only upon the satisfaction of criteria, such as meeting a campaign target [5].

Blockchain-based charitable systems have numerous advantages compared to conventional ones. Initially, they decrease administrative expenses by eliminating intermediaries, so allowing a greater share of gifts to reach the designated recipients. Secondly, they offer improved security via cryptographic protocols, protecting against unauthorized data access and alteration [7]. Ultimately, blockchain's openness can enhance donor involvement by providing verifiable records of donation flows and fund allocation, thereby promoting enduring donor loyalty and support [8].

Incorporating blockchain into nonprofit systems presents hurdles, especially with regulatory compliance, technical complexity, and accessibility. The decentralized characteristics of blockchain may contradict financial regulatory frameworks, which differ by country [6]. Blockchain infrastructure requires technology that may be difficult for smaller organisations with limited resources [10]. Despite these limitations, blockchain's potential to create a reliable and efficient charity system highlights the need for supportive frameworks to promote its adoption.

This paper presents a proposal for a charity system utilizing blockchain technology, with the objective of enhancing transparency, efficiency, and accountability in philanthropic operations. This paper analyses existing blockchain applications and discusses the suggested model's architecture, offering insights into how blockchain might transform the charity sector by addressing present difficulties and creating a new paradigm for donor involvement and trust.

II. LITERATURE REVIEW

Blockchain technology is starting to emerge as a game-changing solution for the charity sector's inefficiencies and trust-related problems. This section offers a critical analysis of blockchain basics, smart contracts, Ethereum-based applications, and the benefits of blockchain in improving transparency, efficiency, and accountability in charitable endeavours. The assessment emphasizes the shortcomings of conventional charity methods and illustrates blockchain's capacity to transform donation management.

A. Blockchain Technology

Blockchain is a decentralized and distributed ledger technology designed to enable secure, transparent, and immutable transactions across a network of nodes. Its immutability ensures that once data is recorded, it cannot be altered, providing a reliable foundation for applications that require high levels of data integrity and security [1]. Each

block in the blockchain contains a timestamp and a reference to the previous block, creating a sequential and traceable record accessible to all network participants. This transparency allows donors to track their contributions, promoting trust and accountability in charitable organizations [2].

One of the key advantages of blockchain is its consensus mechanisms, such as Proof of Work (PoW) and Proof of Stake (PoS), which validate data across decentralized networks without the need for a central authority. This feature is particularly useful in the charity sector, as it reduces the risk of fraud or misallocation through an irreversible and transparent ledger. Additionally, blockchain's decentralized nature eliminates reliance on a single party, reducing vulnerabilities inherent in centralized systems and enhancing security for all stakeholders.

B. Smart Contracts

Smart contracts, which are self-executing agreements with terms directly written into code, automatically carry out actions when specified conditions are met. This makes them particularly suitable for managing charity contributions and ensuring that donor intentions are honoured [4]. In charitable systems, smart contracts can be programmed to release funds only when campaign goals are achieved or to refund donations if targets are not met, providing a dependable and automated solution for managing contributions [5].

Ethereum, the leading blockchain platform for smart contracts, offers a flexible framework for designing and executing these contracts. By embedding specific criteria within Ethereum-based smart contracts, charitable organizations can reduce reliance on intermediaries, streamlining the donation process and enabling efficient fund transfers. A key advantage of smart contracts is their ability to automate complex workflows, such as distributing funds based on milestones, which enhances the security and efficiency of traditional donation systems.

C. Ethereum-Based Applications

Ethereum has emerged as the foundation for decentralised apps (DApps) across multiple sectors, including charity, owing to its powerful smart contract functionalities. The Ethereum platform enables developers to construct decentralised and programmable apps, facilitating tailored solutions for certain use cases, including crowdfunding and donation tracking. The reliability of Ethereum's network and its capacity for complicated decentralized applications provide it an optimal platform for charitable systems necessitating secure, verifiable, and automated processes.

Ethereum-based apps in charity and crowdfunding facilitate transparent gift tracking and milestone-oriented fund distribution. The programmable environment of Ethereum facilitates real-time tracking of gift flows, guaranteeing that money is promptly delivered to beneficiaries and allocated in accordance with donor stipulations. Nonetheless, the scalability constraints and elevated transaction fees of the Ethereum network provide obstacles, especially for minor transactions, potentially impeding its adoption in extensive or micro-donation frameworks [8].

D. Existing Charity Platforms

Traditional crowdfunding and charity platforms, such as GoFundMe and Kickstarter, operate on centralized systems, which inherently limit transparency and control for contributors. These platforms typically act as intermediaries, charging significant fees and holding funds until campaign goals are met. This approach can delay fund disbursement to beneficiaries and raises concerns about accountability, high administrative costs, and the risk of financial mismanagement [9].

Donors frequently lack visibility into the utilisation of their money under a traditional framework, which can breed mistrust and deter further donations. Moreover, centralised control renders donor records susceptible to security threats, increasing the likelihood of unauthorised access or alteration. These constraints highlight the necessity for blockchain's decentralised and transparent methodology, which reduces intermediaries and bolsters the integrity of contribution operations [10].

E. Blockchain-Based Charity Systems

Blockchain-based charitable systems aim to rectify the deficiencies of conventional platforms by utilising the fundamental attributes of blockchain: decentralisation, transparency, and immutability. A blockchain-based charitable system logs each gift transaction onto the distributed ledger, enabling donors and stakeholders to authenticate the allocation of funds. This transparency enhances accountability and confidence, essential for attracting and maintaining contributors [11].

In these systems, smart contracts function as the operational foundation, enabling charities to automate fund disbursements contingent upon certain criteria, such as campaign milestones or predetermined objectives. A smart contract can be programmed to disburse funds only upon achieving a predetermined fundraising target or upon reaching a certain campaign milestone, thus mitigating the risk of financial misappropriation. This automation alleviates the administrative load on charitable organisations and guarantees that donations are utilised as intended, benefiting both donors and beneficiaries [12].

Moreover, blockchain's capacity to generate verifiable, tamper-resistant records improves the auditability of philanthropic transactions, allowing regulatory authorities and independent auditors to examine gift flows reliably. The transparency afforded by blockchain empowers donors by granting them access to real-time data on fund allocation, thereby enhancing their faith in the charitable system [13].

F. Advantages of Blockchain in the Charity Sector

The charity industry can benefit from blockchain technology in several ways, including [20][21] increased operational efficiency, security, and transparency:

1. The public ledger of blockchain enables donors to authenticate fund distribution, hence enhancing confidence and accountability inside philanthropic organisations. This transparency enhances donor involvement and promotes ongoing participation [2, 15].
2. By reducing the necessity for intermediaries, blockchain lowers operational expenses, allowing a larger share of donations to reach recipients.

Conventional methods sometimes incur significant administrative costs, which blockchain technology can markedly decrease [8, 14].

3. Blockchain's cryptographic security safeguards documents from unauthorised access and manipulation. Every transaction is encrypted and authenticated by several nodes, guaranteeing data integrity, particularly in areas with elevated fraud risks [10].
4. Once documented, data on the blockchain remains immutable, increasing confidence in contributors that their donations are safeguarded and will be utilised as intended. The immutability is crucial for establishing a dependable charitable system that promotes trust in donors [9].

According to the literature, blockchain's decentralised, transparent, and safe structure can transform charity systems. By redefining gift management and donor involvement, blockchain addresses insufficient transparency, high administrative costs, and fund misappropriation in traditional charity models. More research is needed to create frameworks for scalable and accessible blockchain integration in the charitable sector due to regulatory compliance and scalability.

III. METHODOLOGY

A blockchain-based charity system architecture using Web 3.0 principles, smart contracts, and decentralised data storage improves transparency, security, and operational efficiency in charitable organisations. The proposed design emphasises smart contract automation to manage donations, fund distributions, and donor goals, contrasting Web 2.0 and Web 3.0 frameworks. System architecture, blockchain integration, and smart contract design are covered here.

A. Comparison of Web 2.0 and Web 3.0 Architectures

In Web 2.0 architecture (Figure 1), a centralised web server handles client requests for the application's front end. After that, these requests reach a database server, which stores all the data, and an application server, which stores the business logic. This centralised structure gives organisations full control over data management, but donors are blind to how their money is used and distributed, limiting transparency.



Fig. 1. System Architecture of Web 2.0

In contrast, the Web 3.0 architecture (Figure 2) integrates blockchain servers and decentralized wallets, enabling secure interactions without intermediaries. Client applications interact with Ethereum blockchain smart contracts using Web3.js calls, while decentralised wallets like MetaMask

authenticate users. All stakeholders can see donation history on an immutable ledger, building trust.

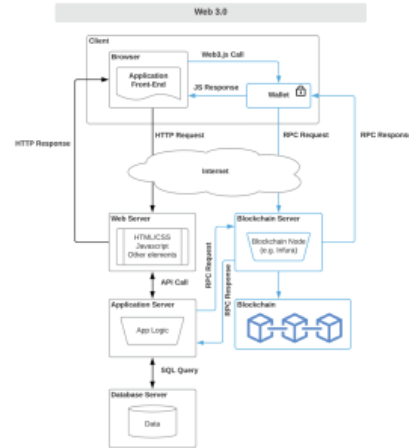


Fig. 2. System Architecture of Web 3.0

B. System Architecture

The architecture of the proposed charity system aims to rectify significant shortcomings in conventional Web 2.0 systems, which are characterised by centralised control of data and transactions, frequently resulting in diminished transparency and elevated operational expenses. Our model shifts to a Web 3.0 architecture, which is fundamentally decentralised, incorporating blockchain to ensure a transparent and immutable record of donations and fund distributions:

1. The Front-End (Integration of React and Web3.js): The client application, constructed with React, incorporates Web3.js to facilitate direct interactions with the Ethereum blockchain. This arrangement enables donors and beneficiaries to access transaction histories and authenticate fund allocations via blockchain records.
2. The donation system is built on the Ethereum blockchain, with smart contracts handling automated fund management. Written in Solidity, these smart contracts are programmed to manage donations based on specific conditions, such as reaching milestones or meeting minimum funding goals. Once deployed, the smart contracts become immutable, ensuring that all donations follow a transparent and rule-based distribution process.
3. For storing large off-chain data, such as donation receipts and beneficiary details, the system uses IPFS. This decentralized storage solution reduces the blockchain's storage load, lowering transaction costs while preserving data accessibility and integrity.
4. To connect to the Ethereum network, the system integrates services like Alchemy and Thirdweb, while MetaMask is used for wallet-based transactions. This setup allows users to complete transactions securely and directly from their wallets, ensuring both a safe donation process and a seamless user experience.

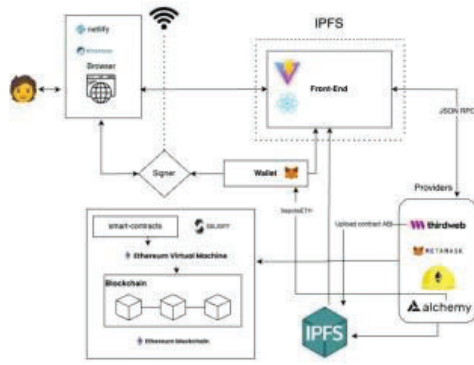


Fig. 3. System Architecture

C. Smart Contract Design and Workflow

Smart contracts are essential for automating fund management and guaranteeing transparency in the charitable sector. Essential processes include:

- Upon the initiation of transactions by donors, smart contracts authenticate the donor's wallet and record each transaction on the blockchain using Web3.js, facilitating real-time interaction and traceability.
- Distribution of funds occurs solely upon the fulfilment of designated conditions, such as achieving a funding milestone. This conditional disbursement improves transparency, enabling donors to monitor fund distribution and campaign advancement.
- If a campaign fails to meet its goals, the smart contract triggers automatic refunds to donors, reinforcing trust by assuring that funds are returned if objectives aren't met.
- All transactions are stored immutably on the blockchain, allowing donors and regulators to audit transaction histories and verify adherence to donor intentions.

The proposed system combines Web 3.0's decentralized structure with smart contract automation to address key challenges in traditional charity models, such as a lack of transparency and the potential misuse of funds. This design allows donors to track their contributions in real time while ensuring that beneficiaries receive funds securely and according to predefined rules. By integrating blockchain technology with IPFS, the system represents a step toward creating a more transparent, secure, and efficient charity platform.

IV. RESULTS

The blockchain-based charity system developed in this study represents a significant step forward compared to traditional crowdfunding and charity platforms, offering enhanced transparency, security, and operational efficiency. By incorporating Web 3.0 principles, smart contracts, and decentralized data storage, the platform overcomes common challenges such as limited transparency, centralized control, and high operational costs. A comparative analysis (Table 1) demonstrates the system's advantages, including transaction-level transparency, automation through smart contracts, and global accessibility—features that set it apart from platforms like GoFundMe and Kickstarter. By utilizing an immutable blockchain ledger, the system ensures a seamless and accountable donation process, providing donors and

organizations with a permanent, verifiable record of all transactions.

TABLE I. COMPARATIVE ANALYSIS

Feature	Platforms		
	Proposed Platform	GoFundMe	Kickstarter
Transparency	Real-time tracking of fund allocation is available to all stakeholders.	Donors rely on platform-provided updates without independent verification.	Moderate transparency; milestone-based updates are platform-controlled.
Fund Distribution	Smart contract automated	Manual fund management.	Funds are distributed manually.
Fees	Minimal gas fees	2.9% + \$0.30 per donation	5% + payment fees
Accountability	Immutable records ensure full accountability for donations.	Trust depends on platform credibility and creator reliability.	Trust based on platform's creator policies but lacks external auditability.
Refunds	Automated refunds if campaign goals are not met, handled by smart contracts.	Refunds depend on platform and creator policies.	Refunds are initiated only for unsuccessful campaigns.
Global Accessibility	Accessible to anyone with a blockchain wallet like MetaMask.	Regional restrictions may apply.	Restricted to selected countries.

The Home Page (Figure 4) provides an interactive platform by depicting and providing explanations of the different charities on offer. The system seeks to increase donor participation by embedding appealing and relevant information, thus helping in creating an appropriate initial stage for the users wishing to look for charity opportunities.

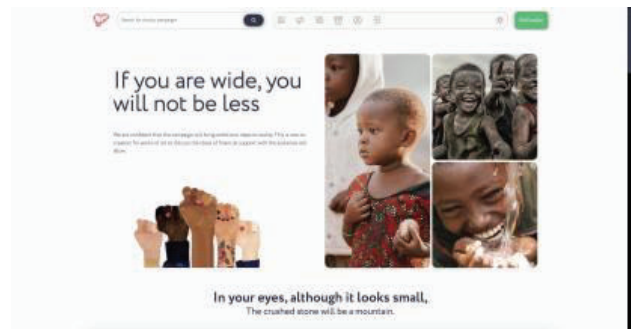


Fig. 4. Home page

The Create Campaign Page (Figure 5) offers a user-friendly, intuitive, and comprehensive interface for creating new campaigns. Users can easily input key details such as the campaign title, description, fundraising goal, and deadline[19]. However, it's important to note that Sepolia ETH must be used during campaign creation. This requirement ensures that campaigns are created on the Ethereum network, where each campaign is linked to a smart contract. These smart contracts provide a secure and transparent record of how funds are managed and spent.

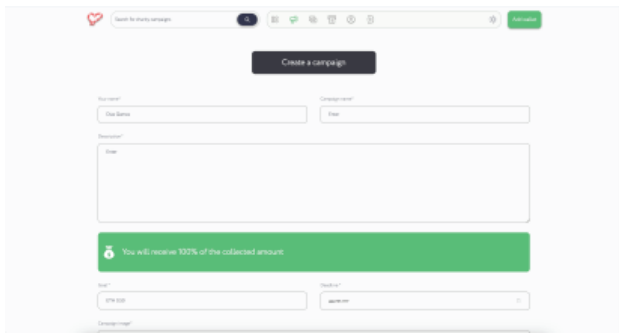


Fig. 5. Create campaign page

The Campaign Details Page (Figure 6) gives contributors a clear view of key campaign statistics, such as the total amount raised, the number of days remaining, and the number of contributors[18]. This page plays a crucial role in helping donors track their contributions and stay informed about the campaign's progress. It also builds trust by providing transparency, as donors can see how funds are being managed and used in real time.

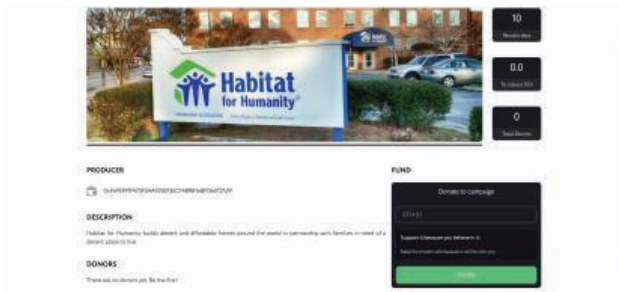


Fig. 6. Campaign details page

Authentication is managed through MetaMask, a decentralized wallet extension that ensures users can securely connect to the platform. With MetaMask, users can approve transactions without exposing sensitive personal information, providing an extra layer of security and privacy during the donation process. The donation process itself is smooth and straightforward, as shown in the MetaMask Wallet Confirmation (Figure 7), making it easy for users to contribute with confidence.

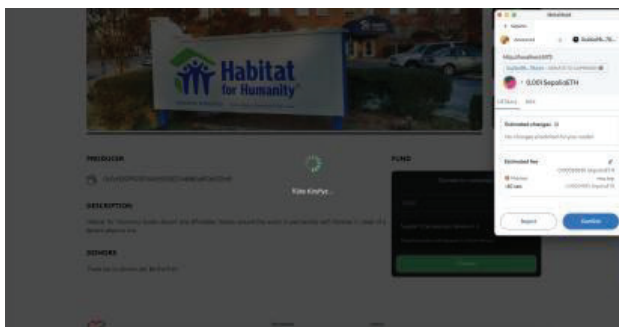


Fig. 7. MetaMask Wallet Confirmation

Donors use MetaMask to confirm their contributions, and once approved, the transaction is recorded on the blockchain. The Donation Transaction Success Screen (Figure 8) provides instant confirmation, giving donors peace of mind and

ensuring transparency by showing that their donation has been securely processed.

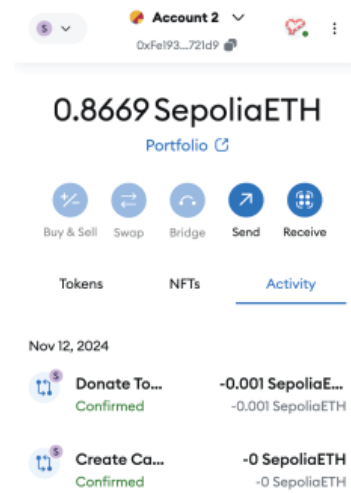


Fig. 8. Donation Transaction Success Screen

Finally, the blockchain's inherent auditability ensured compliance with regulatory standards, as each transaction was verifiable and accessible to external parties. With its immutable, decentralized ledger, the system created a reliable audit trail, allowing regulatory bodies and independent auditors to confirm that funds were used as intended. This level of transparency and accountability set the system apart from traditional donation platforms, which often lack such assurances.

By leveraging blockchain technology, the charity system effectively addressed major shortcomings of conventional charity models. Features like smart contracts, decentralized data storage, and wallet-based authentication provided greater transparency, security, and automation. The user interface, highlighted through screenshots, showcased its simplicity and accessibility, empowering donors and beneficiaries to engage in charitable activities with confidence and ease.

V. CONCLUSION

This study highlights the transformative potential of blockchain technology to revolutionize the charity sector by addressing key challenges such as lack of transparency, high administrative costs, and fund mismanagement. Leveraging blockchain's decentralization, immutability, and cryptographic security, the proposed system enhances efficiency and accountability in charitable operations.

Smart contracts distribute funds and enforce donor conditions, eliminating intermediaries and lowering operational costs. Donors trust immutable smart contracts for secure, transparent transactions. IPFS decentralises campaign image and document storage, improving scalability, resource efficiency, and data integrity. Wallet-based authentication, like MetaMask, lets donors approve transactions without sharing personal data, following blockchain's privacy and security principles.

Future work could explore mobile app accessibility, multi-blockchain compatibility to reduce transaction fees, and advanced analytics to enhance donor insights. These advancements could improve usability and scalability, broadening the platform's appeal.

In conclusion, the blockchain-based charity system offers a practical and scientifically grounded solution to longstanding issues in the charity sector. By creating a transparent, secure, and decentralized donation platform, it can reshape charitable giving, ensuring contributions are managed with integrity and trust.

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Comparative Analysis of Blockchain Platforms for Charitable Donation

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Abstract— In the COVID-19 era, charity applications have become the need of the hour through which such opportunities have provided hassle-free and remote donation to the needed people. However, this growing adoption raises concerns about transparency, security, and the mismanagement of donated funds. While blockchain technology holds great promise, it also faces challenges such as high costs, energy consumption, and scalability limitations. These challenges highlight the need to identify the most suitable blockchain platform for charity applications and optimize their performance. Even though numerous platforms constructed their blockchain applications based on Ethereum as it was the first blockchain with smart contract capability, newer alternatives such as Solana, Tezos, Polygon, and Binance Smart Chain come with their own advantages and disadvantages. Some of the key points used in the evaluation are transaction cost, transaction speed, security & decentralization. Solana stands out particularly in terms of speed, costs, and scalability- with 65,000 transactions per second through the Proof of History mechanism. Polygon offers low costs with great scalability by using all the benefits of the Ethereum ecosystem as well as layer-2 solutions, other advantages from Tezos and Binance Smart Chain are decentralization and the system integrity, even though the whole system lags behind in speed and cost- effectiveness. Based on performance and development environment, Solana (with Rust) and Polygon (with Solidity) emerge as leading choices for building modern charity applications. This article provides insights for developers and organizations interested in building strong, transparent, and efficient systems that tackle global issues in philanthropy.

Keywords—Blockchain, Charity Applications, Solana, Ethereum, Polygon, Binance Smart Chain, Tezos.

I. INTRODUCTION

Blockchain has innovatively transformed the finance world, through decentralization and security on transactions. Hence, it is a worthwhile candidate for charitable donation applications. Traditional charity systems mostly lack transparency, hence the mistrust fostered by malpractices. Blockchain addresses this by enhancing transparency and ensuring secure utilization of donated

funds. Utilization of blockchain will enable donors to trace donations, elevating their confidence in the charity system. With these benefits, blockchain integration has some other disadvantages, such as transaction costs, speed, and scalability, which require caution. To ensure the charity application, an effective choice of the proper blockchain must be made. Among various blockchains, five prominent ones, namely Ethereum, Tezos, Solana, Polygon, and Binance Smart Chain, have been highlighted as capable of successfully executing smart contracts and having decentralized applications. This comparison analysis of the blockchains analyzes and evaluates each one based on key parameters such as transaction fees, speed, security and decentralization. Ethereum has great security and widespread adoption but still shares high fees of transactions. Tezos promises to be energy efficient and self-amending, hence sustainable. Solana boasts very high throughput with low transaction cost, whereas Polygon balances cost-effectiveness with Ethereum compatibility; however, Binance Smart Chain boasts affordability with fast processing but pulls at decentralization concerns. This review shall identify the blockchain that best fits the charitable donation application but should be a balance of transparency, efficiency, and trustworthiness. Conclusion derived from findings would then enlighten how to effectively make use of blockchain for philanthropy.

II. LITERATURE REVIEW

Charity Wall, using the Ethereum and VeChain blockchains with PoS and PoA algorithms respectively, explains how blockchain ensures secure tracing and notarization of donations so that the application eliminates intermediaries and reduces fraud risks, and ensures funds utilized do so as intended[1]. UpHaaR scheme further extends the study on the application of blockchain technology for transparency in donations. It used two blockchain technologies: Ethereum and VeChain through Certificate-less Cryptography (CLC) and Certificate-less Signcryption Scheme (CSS), which proved how blockchain technology

maintains tamper-proof, untampered records that boost donor's confidence [2].

Being able to put in smart contracts has been one of the flagship features of many blockchain-based systems, as seen in the Fundraising Tracking System. It was originally based on Proof of Work for reaching consensus. However, Ethereum has switched to Proof of Stake, which solved concerns with high energy consumption while further securing transactions. However, high computational costs as well as scalability issues remain significant open challenges concerning the actual deployment of blockchain systems [3]. Blockchain applications have entered niche segments like charity auction NFT markets. For instance, the authentication of NFTs in the Fisco Bcos blockchain employs multi-signature algorithms and Elliptic Curve Digital Signature Algorithm (ECDSA) for transaction integrity in a safe manner[4].

On the contrary, Ethereum-based traceability platforms based on smart contracts in Solidity makes public-permissioned blockchains aimed at a higher transparency with barriers still presented by expensive mining operations and ensuing costs[5]. The other quite popular blockchain, Solana, integrates Proof of History (PoH) with Proof of Stake (PoS) and consequently achieves an extremely high throughput by handling up to 65,000 transactions per second. For this, traceability of products in supply chains is a fairly decent application in real-time that Solana portrays. It enables the tracking of its products through IoT devices like Sigfox cloud gateways. Being scalable, it also boasts of cost-effectiveness, thereby presenting an alternative solution. However, consumption of energy and operational costs remains an area of concern for widespread adoption[7] [8][9]. Tezos brings to use a self-amending PoS mechanism with no hard forks, ensuring easy governance on-chain. Other applications include document tracking systems which achieve high precision and efficiency mainly with issues associated with balancing on-chain and off-chain computational loads[11][12].

Comparative studies also highlight the growth of other blockchains' participation, for example, Polygon and Binance Smart Chain (BSC), in addressing the various challenges. For instance, Polygon provides a layer-2 solution that provides a scalable decentralized framework to handle digital evidence with integrity and transparent assurance through the capabilities of its smart contract functionality[15]. Meanwhile, BSC is applied to the medical record, enabling decentralized systems to store atomic transaction records securely and promote the ease and availability of patient data[17]. Such innovations epitomize the flexibility of blockchain in solving domain-specific problems even at the risk of financial and technical barriers. Such applications of blockchain are above charity, like fraud detection, content tracking, and governance, in a decentralized system.

Ethereum is known for high throughput as well as functionality in the execution of smart contracts. Governance and transaction integrity are the key values of Tezos and Fisco Bcos. Even at its transformatory strengths, blockchain still needs to overcome quite a number of difficulties relating to regulation, an exorbitant energy requirement, low technological acceptance, and financial costs. Much will be

done to get it spread across applications that thrive in transparency, security, and trust[6][8][15][17].

From the above studies, I observed that blockchain for smart contract applications are influenced by factors of a transaction cost, transaction rate, security and decentralisation, scalability issues which is also key components for charitable application using blockchain, from the study that I observed that solana provides high transaction throughput and less transaction cost and tezos gives on chain governance which promotes the decentralisation and polygon uses the existing features of ethereum while reducing the transaction cost and binance smart chain provides high transaction speed at low and also it is compatible with the famous ethereum blockchain, while ethereum moving towards proof of stake consensus for reducing computation consumption. Although the studies represented high usage of the blockchain technology, I am going to use the studies in the charity donation application and this paper gives a profound understanding and allows one to choose the blockchain based on the criteria, such as transaction cost, transaction speed, security & decentralization that enhances charitable donation application performance.

III. BLOCKCHAIN COMPARISON

In this article, I intend to compare the blockchain based on the criteria like Transaction cost, Transaction speed, Security and Decentralization, and Scalability in the end which allows us to choose a blockchain that suits the requirements of charity applications

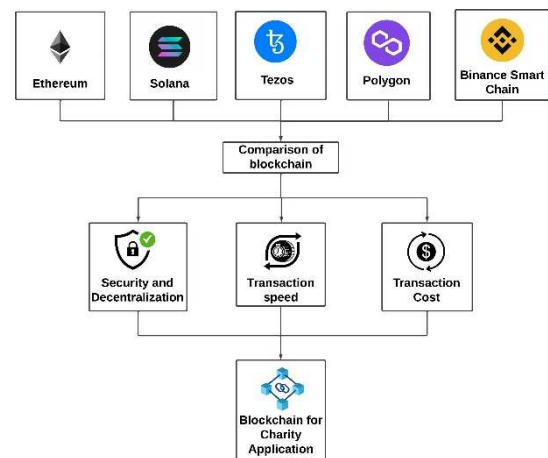


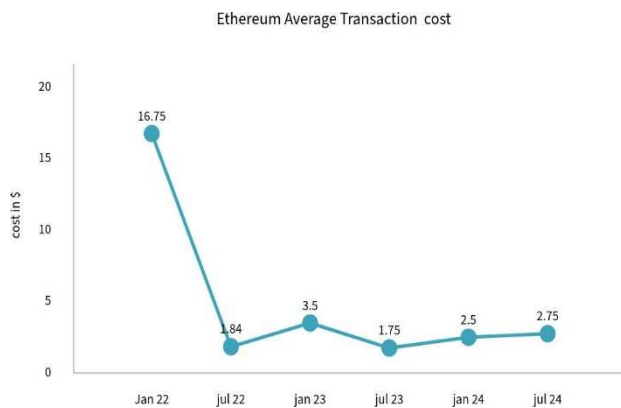
Fig. 1. Blockchain Comparison

A. Transaction Cost

Transaction cost is an important criteria for blockchain based charity applications, because when the transaction cost are high, it might promote negative impact in the small donation which impact the overall donations. When the transaction cost are minimal it promotes everyone to donate irrespective of the amount that is donated. Hence Transaction cost is an important aspect in blockchain based charity applications.

1) Ethereum

The cost of performing a transaction on the Ethereum blockchain is gas fees, paid in ETH. These fees include a base fee, required to process, and a priority fee, an



optional "tip" to be processed faster; the base fee was adjusted with network congestion and burned according to EIP-1559, whereas the priority fee incentivizes the validators. The gas fee formula is, $\text{Gas Fee} = \text{Units of Gas Used} \times (\text{Base Fee} + \text{Priority Fee})$. Gas fees are influenced by network demand, congestion, and transaction complexity[24]. Simpler transactions like ETH transfers cost less gas, while executing dApps or complex smart contracts requires more[18].

Fig. 2. Ethereum Average Transaction cost

2) Solana

Solana's transaction costs are minimal, averaging \$0.00025 per transaction, with high throughput enabled by Proof of History and parallel processing. Fees are calculated by the base fee - per signature -and the resource consumption cost, which is compute units, CU. The formula for transaction fee is, $\text{Transaction Fee} = (\text{Base Fee} \times \text{Signatures}) + (\text{CU Consumed} \times \text{CU Price})$. Solana also allows priority fees to accelerate transactions. A portion of fees is burned, reducing SOL supply, while the rest rewards validators[19]. Though stable, the static base fee can lead to block saturation, affecting transaction costs during congestion.

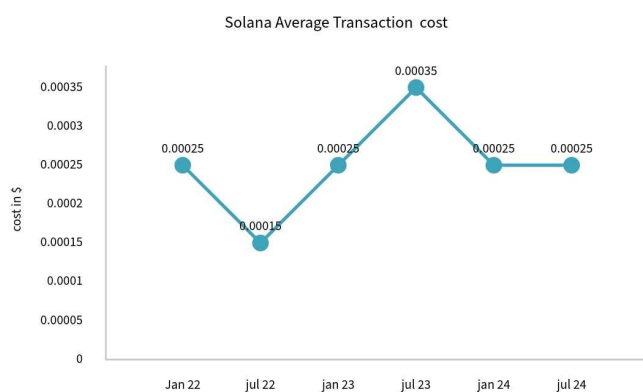


Fig. 3. Solana Average Transaction Cost

3) Tezos

Tezos operates using a native cryptocurrency, XTZ (tez), which pays the cost for transactions, thereby keeping fees low for access. The fee is a combination of Basic Fee, which is set, and a Gas Fee, which is an expression of computational effort. The formula is: $\text{Transaction Cost} =$

$(\text{Gas Consumed} \times \text{Gas Price})$ Each block has a 10 million gas unit cap, while individual transactions have a gas unit cap of 1.04 million. A charge is made based on data size in storing fees, which promotes efficiency, while in burn fees, XTZ supply is reduced. Fees dynamically adjust based on network demand as well as transaction complexity. Baking or endorsement is useful to a user but involves depositing a certain amount of Tz as security for a certain number of validated blocks. In case double baking or endorsement occurs, the platform slashes the security deposit[21].



Fig. 3. Tezos Average Transaction cost

4) Polygon

Polygon is a Layer 2 scaling solution for Ethereum, providing a much better performance through cheaper fees and high throughput[22]. Calculation of the transaction costs would be as follows: $\text{Transaction Cost in USD} = \text{Transaction Cost in MATIC} \times \text{Price of MATIC at the moment}$. Fees are ranged at an average \$0.0005 to \$0.002 per transaction because of its use of Proof of Stake and Plasma framework, supporting up to 65,000 TPS. Polygon comprises the two-layer structure of the security of Layer-1 of Ethereum and scalability on Layer-2 with zkEVM supporting fee accuracy. Gas prices depend on network congestion, complexity of the transaction, and mechanisms of EIP-1559. Fees are generally found below \$0.01.

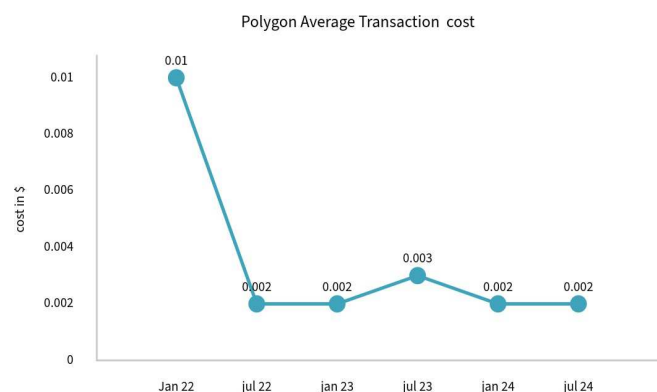


Fig. 5. Polygon Average Transaction cost

5) Binance Smart Chain

The Binance Smart Chain uses the PoSA consensus algorithm and charges gas in Binance Coin (BNB). The fees for transactions are, $\text{Transaction Fee} = \text{Gas Used} \times \text{Gas Price}$. The gas prices range between 5 gwei and 20 gwei, based on

the activity in the network. Currently, the average price of a single transaction is \$0.10. Gas fees incentivize validators, give a priority to transactions in time of congestion, and prevent spam. Lower costs come with simpler token transfers, while complex smart contract activities require more gas. Due to this efficient design, BSC is both affordable and scalable [23], supporting a variety of applications with users encouraged by low fees for adoption and performance.

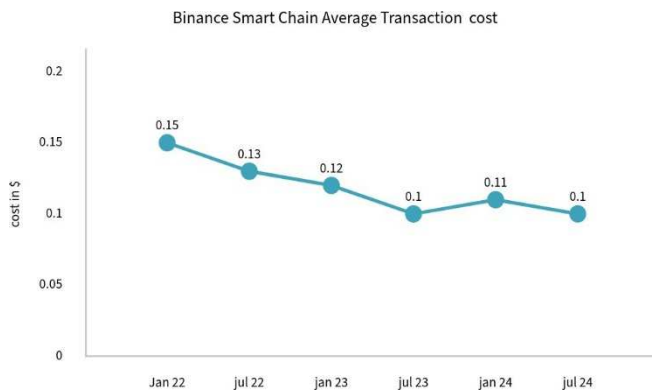


Fig. 6. Binance Smart Chain Average Transaction cost

Looking at each blockchain's transaction cost structure reveals some insight into gas fee drivers and how networks structure transaction costs. For instance, average fees for simple transactions on Ethereum are \$1–\$5, but gas-intensive smart contracts can easily exceed \$20+ during congestion. Meanwhile, its efficient Proof of History (PoH) mechanism has kept the average transaction cost on Solana at a lowly \$0.00025. The average transaction fee on Tezos is around \$0.01–\$0.05 and depends on their computational complexity. Polygon costs \$0.002–\$0.01 on average, taking advantage of the low-cost side-chain layer that builds off Ethereum compatibility, and BSC costs a whole \$0.05–\$0.10 per transaction based on its dual-chain infrastructure. Compared to everything on this list, here, the ultimate winner is Solana with ultra low transaction fees.

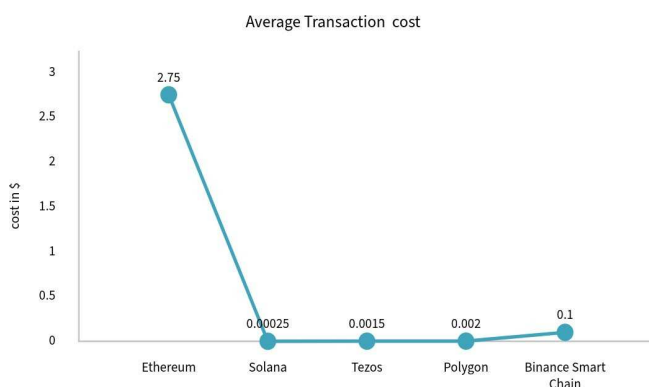


Fig. 7. Average Transaction cost

B. Transaction Speed

Another very crucial comparative aspect concerning blockchains is the speed of transactions, especially in emergencies like the COVID-19 pandemic. The faster the transaction, the faster the donation will hit the charities and the funds used for the distressed people. In such an

emergency as this, even a minimal delay in receiving donations may halt relief activities. Secondly, the accelerations in the transaction speeds advance the general experience of the donor, and consequently, they end up having more involvement in philanthropy. With high-speed transactions that assure funds being transferred without any impediments and within appropriate time, this creates trust and boosts donations. A comparison based on the speed of blockchain transactions can be very enlightening on whether they are applicable for charity-based purposes.

1) Ethereum

The PoS protocol in Ethereum processes transactions at a rate of 15-30 TPS. In its raw form, it lags behind newer blockchains, but it still enjoys major improvements through Layer 2 solutions like Optimistic and ZK-Rollups [25] that package several transactions offchain, reducing congestion and improving throughput, especially during an event. While this technology brings with it some slowdowns from network congestion and occasionally sky-high gas fees, upgrades like shard chains promise to bring Ethereum into the thousands of TPS in the near future, thus making it highly feasible for time-sensitive applications like charitable donations.

2) Solana

As for transaction speed, Solana takes the cake with up to 65,000 transactions per second, thanks in part to innovation in Proof of History (PoH) and Sealevel parallel processing. These help to enable parallel transaction processing within the network and quick validation of these transactions at an approximately 400-milliseconds block times rate [20]. Another protocol known as its Gulf Stream helps to reduce latency by pre-forwarding transactions to validators for quick validation. These make it support high-speed application processing where transactions have to occur in near instantaneous time, like real-time relief efforts during emergencies, but Solana has suffered with traffic spikes that sometimes have been known to degrade its performance occasionally.

3) Tezos

Tezos accommodates a block time of 30 seconds, with Liquid Proof of Stake mechanism supporting transaction speeds of 34 TPS. Its modular structure allows upgrades that augment speed and scalability without affecting the network. Still, smart contracts may hinder unlocking the full potential of high throughput; rollups can help offload transaction processing so efficiency is fully realized when demand peaks. Tezos' governance model allows for timely congestion relief updates but is still slower than super-fast blockchains like Solana and Polygon.

4) Polygon

Plasma chains and rollup technologies such as zk-rollups and optimistic rollups help Polygon hit a jaw-dropping TPS speeds of up to 65,000. These technologies batch off-chain transactions while periodically committing them to the main Ethereum network; latency and costs are, therefore reduced. The network architecture, which includes Bor and Heimdall, ensures quick creation and validation of blocks. High congestion events will always cause temporary

slowdowns, but the subtle optimizations go with the name Polygon Zero. Polygon is fast and compatible with the Ethereum[27], making it appropriate for high-performance charitable applications.

5) Binance Smart Chain

By achieving a transaction speed of 140 TPS, with block times under 3 seconds, BSC approaches the balance point between scalability and decentralization through its Proof-of-Staked-Authority mechanism. Developers can also migrate dApps easily due to compatibility with Ethereum, and further scaling solutions such as opBNB Layer 2 rollups. BSC has seen low fees and high speeds, [29] so DeFi and NFT-related platforms are utilizing it, although the number of validators is a bit small concerning centralization. Its efficiency and ability to handle millions of transactions every day make it a strong candidate for charity-based applications that require fast transfers of funds.

The transaction speeds and throughputs achieved by a blockchain expose the layers that may reach the highest throughputs to support blockchain-based applications in charitable work. At the lowest transaction rate, Ethereum only offers 15–30 TPS, thus being heavily constrained by such high-demand conditions. Tezos offers a slight bump at 40 TPS, though Binance Smart Chain currently boasts a far more substantial advantage over Ethereum at 140 TPS. This last two are notable, though, in terms of really incredible transaction speeds that can reach up to 65,000 TPS, making them the optimal choices for those applications requiring high throughput. Solana and Polygon are nice for charity applications, with high-speed transactions, however, when it comes to fast-paced operations, they have a great advantage.

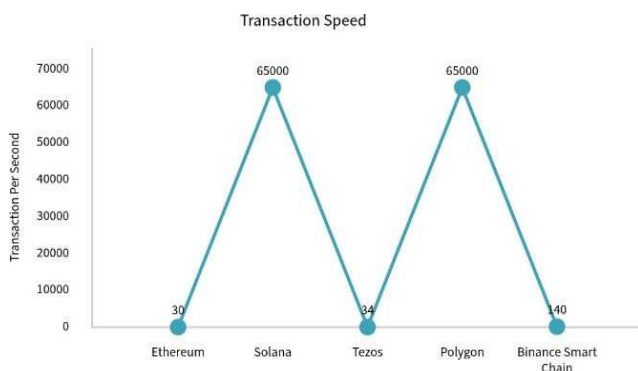


Fig. 8. Average Transaction Speed of the blockchains

C. Security and Decentralization

While choosing a good blockchain for charity applications, security and decentralization represent crucial parameters. These two aspects form the backbone of an organizational structure within the charity sector by promoting trust and reliability. A blockchain that offers security and decentralization relates to a transparent nature wherein donors can trace their contributions and verify the allocation of funds. Transparency assures minimal misuse of contributions therefore maximizing public trust in charitable

organizations, as well as donations in volumes. Security guards sensitive donor information as well as transaction data, hence minimizing fraud, as well as cyberattacks. Decentralization involves the removal of middlemen to ensure the finances reach their destination effectively.

1) Ethereum

Although the Ethereum has a massive sense of decentralization and security, it is truly an attractive platform that makes it suitable for charitable applications. It has been shifted from PoW to PoS, hence it becomes accessible in order to build broader participation with fair control by a greater number of validators. More scalability is added through sharding, where the blockchain is broken into pieces, cutting bottlenecks, and ensures resilience in the network. This coupling of the Gasper consensus mechanism-Casper FFG, GHOST-helps prevent Sybil attacks as well as avalanche attacks [29], and PoS validators are incentivized to act honestly with economic penalties. This will ensure transparent, tamper-proof donation records.

2) Solana

Combining Proof of History with Proof of Stake, Solana achieves very high throughput: averaging at 2,812 TPS[20]. Thus, its network can be described as ideal for high-speed applications, but the visibility of the relatively small validator set due to demanding hardware raises decentralization concerns. Ethereum boasts over 800,000 validators, which represents a more decentralized network; however, the scalability issues lead to a relatively higher cost or fees and slower transactions during peak demand times. Rollups are great at alleviating congestion. Such Layer 2 scaling solutions; Solana gives stiff cryptographic timestamps but suffers downtime under high loads. It has a matured smart contract ecosystem which promises secure automatic transactions with the support of complex decentralized applications.

3) Tezos

Tezos is decentralized and secure through self-amendment and on-chain governance [11] that lock in hard forks while guaranteeing the stability of the network. Its decentralized mechanism verifies transactions without recourse to authority, a neutral core ability of the blockchain technology. Tezos approaches the trilemma of decentralization, scalability, and security through innovative mechanisms for consensus. Smart contracts grounded in formal verification mathematically guarantee the correctness of transactions for enhancing operational security. Its governance model formalizes voting for protocol changes, eliminating conflicts and enhancing security through stakeholder participation. This approach further tightens the network's response to threats and innovations while being theoretically secure due to its self-amending ledger and formal verification. However, Tezos, like any other decentralized network, requires vigilant implementation to avoid decentralization-induced vulnerabilities.

4) Polygon

Polygon blockchain is safe and decentralized because of its robust architecture as well as by the consensus mechanism, Proof-of-Stake. In a network, validators need to stake MATIC tokens, which encourages good behavior and punishes malicious ones, thereby improving the security of the network. All the scaling solutions offered by Polygon at Layer-2, such as Plasma Chains and commit chains of Polygon PoS, benefit from all the security features of Ethereum by processing transactions off-chain to increase scalability without trading in decentralization[27]. Decentralized governance by Polygon makes it possible for stakeholders to participate in key decision-making processes, thus working towards an inclusive approach of minimizing central points of failure. In addition, it further supports security with a modular framework, so developers can create custom blockchain solutions for specific use cases without losing out on the full potential of the underlying security model of Polygon.

5) Binance Smart Chain

The BSC is making frictionless with decentralization through scalability with its Delegated Proof

of Stake consensus built on 21 validators[16], making transactions faster and less expensive. Increased risk for centralization coupled with a probability of collusion in broken validators are the worries, though partial decentralization is preserved through cross-chain Ethereum compatibility, the smooth process of migration of projects. Its layered architecture dissects transaction validation from smart contract execution, which further improves performance with little sacrifice in the name of security. With low transaction costs and high throughput, BSC has been pretty well known for decentralized applications, but its centralized nature poses increased risk through attacks.

Among the blockchains, what Ethereum offers is a mature ecosystem and a large validator network next to decentralized governance, making it one of the blockchains with the highest levels of trust and transparency, with mission-critical applications. Solana makes good based on speed but fails hard on centralization. Polygon and BSC scale at the expense of true decentralization. So far, as reliable as Ethereum is to secure, transparent, and trustless charity platforms.

TABLE I. COMPARISON OF FIVE BLOCKCHAINS ON THE EVALUATION CRITERIA

Evaluation Criteria	Ethereum	Solana	Tezos	Polygon	Binance Smart Chain
Transaction cost	\$2.75	\$0.00025	\$0.0015	\$0.002	\$0.1
Transaction speed	30 TPS	65000 TPS	34 TPS	65000 TPS	140 TPS
Consensus mechanism	Proof of Stake (after Ethereum 2.0)	Proof of History + Proof of Stake Hybrid	Liquid Proof of Stake	Proof of Stake	Proof of Staked Authority (PoSA)
Security	High (Extensively tested smart contracts)	High (Advanced cryptography and consensus)	High (Formal verification ensures reliability)	High (Backed by Ethereum security)	Moderate (more centralized raising concerns)
Decentralization	Moderate (Highly distributed but limited nodes)	Moderate (Validator concentration risk)	High (Inclusive governance model)	Moderate (Dependent on Ethereum's decentralization)	Moderate (Less decentralized due to validator count)
Scalability	Limited (High congestion on base layer)	High (Optimized for high throughput)	Moderate (Scalable for moderate usage)	High (Scales effectively with Layer-2 solutions)	High (Optimized for DApps and fast transactions)

IV. SUMMARY AND FINDINGS

On the basis of the comparison of Ethereum, Solana, Tezos, Polygon, and BSC, their merits and demerits of using blockchain-based charity applications would be obtained. Based on transaction cost criteria, solana has the lowest at \$0.00025 and highest transaction cost at \$2.75 by Ethereum, which leads to be cost efficient as the primary objective of the project. Transaction speeds output Similar:

Solana and Polygon take the lead at 65,000 TPS, while Ethereum and Tezos is well below 50 TPS, respectively. As far as the consensus mechanisms are concerned, Ethereum (Proof of Stake), Tezos (Liquid Proof of Stake) and Polygon provides energy efficiency while Solana and BSC present moderate efficiencies. Security is very strong across most blockchains with case in point being

extensively tested protocols applied by Ethereum and Tezos, though the BSC's nature being centralized does raise concerns. Decentralization is high in Tezos compared to other blockchains, which is inclusive governance-wise, while Solana and BSC are semi-decentralized due to the concentration of validators. Scalability is high for Solana, Polygon, and BSC, which have efficient throughput as well as Layer-2 solutions, and there are issues with congestion for Ethereum. In brief, Tezos presents a balanced decentralization, security, and energy efficiency; Solana and Polygon outshine other blockchains for its scalability and pace. The security is better for Ethereum; however, in terms of scalability and cost, BSC has solutions that are fast and cost-effective but not exactly decentralized.

V. FUTURE SCOPE

The review article can be enhanced in several ways to obtain a more comprehensive understanding of blockchain application for charitable organizations. Some future research might be carried out on real-world case studies, with suitable charity applications built on blockchains, such as on Solana, Tezos, and others. An evaluation of their implementation strategies, difficulties involved, and outcomes might provide practical insights. Further, including emerging blockchains such as Avalanche or Cardano in the comparison will depict a more inclusive view of the blockchain landscape while it is still in its evolution and therefore not disregarded by time.

VI. CONCLUSION

This review article evaluates the suitability of Ethereum, Solana, Tezos, Polygon, and Binance Smart Chain for blockchain-based charitable applications by considering factors such as transaction cost, speed, consensus mechanisms, energy efficiency, security, decentralization, and scalability. Of these, Solana is the most suitable blockchain for charitable applications, with its high transaction throughput, very low transaction costs, and ability to process micro-donations. These features ensure smooth operations even under a high network demand, thereby ensuring it is highly scalable as well as cost-effective. Other blockchains, like Tezos, provide other valuable benefits, such as energy efficiency and a strong sense of governance. However, Solana's emphasis on speed and affordability matches the needs of charity applications closely, where microtransactions are more common and the potential users are so much broader. The booming community and support of decentralized applications on Solana also contribute to the strength of this innovation platform in the nonprofit sector. The case for recommending Solana is, therefore, how it can enhance transparency and trust while boosting efficiency in charitable action. Future work might involve hybrid models or integrations with Layer-2 solutions to further optimize blockchain applications for philanthropic purposes, ensuring greater impact and inclusivity in global charity efforts.

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Blockchain-Based Charitable Donation Privacy Protection Scheme

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Abstract

Blockchain has the characteristics of decentralization, tamper-proof, information traceability, openness and transparency, and has a good fit with charitable donations. Aiming at the problems of data center management, poor transparency and tampering in the existing charity traceability process, a secure sharing scheme of charity donation data based on blockchain is proposed. The scheme uses the blockchain to store the hash value, key and access policy of the encrypted data, and the original data is stored in IPFS(InterPlanetary File System), which ensures the privacy of users. A trust model based on reward and punishment mechanism is proposed to evaluate the trust value according to the behavior of nodes. Ciphertext policy attribute encryption technology combining trust and time attribute is used in smart contract to design data security sharing method. After evaluation and analysis, the scheme has the characteristics of openness and transparency, decentralization, and difficulty in tampering, which provides a new reference for solving the long-standing corruption problem in the charity field.

Keywords

blockchain,charity,access control

1 Introduction

In modern society, various charitable forces represented by charitable organizations have grown rapidly, and social charitable awareness has been significantly enhanced. Charitable donations through charitable organizations play an important role in national and regional responses to emergencies, such as the Novel Coronavirus (Covid-19) epidemic, earthquakes, typhoons, floods and other disasters, as well as social and public issues such as poverty alleviation, pensions, and diseases. With the disclosure of various charitable scandals, problems in the process of charitable donations have been exposed to the public one by one, such as the lack of credibility of charitable organizations, corruption of charitable organizations, low transparency of information disclosure, and the inability to trace the use of donations, which has led to scandals in the charitable industry. The public gradually lost trust in the charity industry.

The emergence of blockchain brings hope to solve problems in the field of charity, everyone turns their attention to promoting the transformation of public welfare into blockchain, hoping to use blockchain to realize on-chain donations and information traceability. The features of blockchain, such as decentralization, openness and transparency, information that cannot be tampered with and traceable, and smart contracts, can perfectly solve the problems in philanthropy. Combining blockchain technology with the public welfare system can effectively monitor the whereabouts of each donation, forming a closed loop from fundraising to donation. Due to the characteristics of blockchain technology, it is difficult to modify the data once recorded, which can greatly improve the public's trust in non-profit organizations, so that non-profit organizations can self-certify their fairness and fairness, improve transparency, and build credibility.

In recent years, academics from around the world have performed substantial research on how to address the problem of public distrust in philanthropic organizations. Basil D Z summary how might

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cause-related marketing affect attitudes toward the charity involved[1]; MD Diarmuid Analyzed the characteristics of charities in charities to drive charitable accountability [2]; Inger Lyngdrup explores when and why charity vanished from public relief in Copenhagen and thereby differentiates itself from the traditional research field on welfare that focuses on studying the growth of taxation and social spending in the 19th century[3]; Davidson S points to potential fraud by middlemen who oversee and manage money[4]; A. Farsya Kirana aims to examine the institutional mechanism and information systems success factors that influence trust and distrust in the online donation platform and how it influences attitude and online donation intention[5]. Some scholars tries to propose a charitable donation system combined with Internet technology, Hai-ying YU proposes a model of trust formation in online charity information, rooted in the information processing theory [6]; Lanerolle proposes a system that manages donations after investigating current disaster management system in Sri Lanka and in several other countries which get affected by natural disasters frequently [7]. D.Chhibber Virtualizing Food Donation Distribution through Mobile Application and Cloud-Based Supply Chain Management[8]. Although the above methods have certain improvement effects, they have not changed the problems existing in the traditional charitable donation process, and the credibility of charitable undertakings is still low. With the development of blockchain technology, a type of solution has emerged, namely blockchain + charity: Mohd Nor R proposed a decentralized, authentic and transparent donation system to solve the problems of additional fees, information opacity and processing delays in online donation platforms[9]. Wu H developing a reliable service system of charity donation loaded on the blockchain to cope with the complex service needs of charities due to the COVID-19 epidemic[10];Farooq M S designed a blockchain-based charity management platform aiming to provide a transparent, secure, auditable and efficient system[11].

These methods make some improvements in the donation method, but they do not consider the privacy information involved in the donation process, and the privacy information of participants is not protected. We think attribute encryption can be used to protect privacy. Zhang X proposed a novel blockchain-based architecture for data sharing with attribute-based cryptosystem (BaDS), the architecture can achieve privacy-preserving, user-self-controlled data sharing, and decentralization by using blockchain and several attribute-based cryptosystems[12]. Axin Wu In view of the efficiency of ABE, the privacy protection of the attribute and the abuse of the secret key, the efficient and privacy-preserving traceable attribute-based encryption in blockchain is proposed[13]. Zhang X propose a privacy-preserving and user-controlled architecture for data sharing based on blockchain system and Ciphertext-Policy Attribute-Based Encryption (CP-ABE), called ThemisABE[14].

In this paper, we propose a charity foundation scheme based on blockchain, Our main contributions are as follow:

- We propose a blockchain-based charity donation privacy protection scheme that aims to provide a transparent, secure, auditable, and efficient system. At the same time, we established a trust model for nodes, and a trust evaluation framework is proposed.
- Combining symmetric encryption with attribute encryption, we not only protect the confidentiality of data, but also provide fine-grained access to data according to user roles and attributes.
- We use the time attribute to automatically revoke the user's attribute key, which solves the key management problem existing in the ABE solution.

2 Preliminaries

2.1 Blockchain

On October 31, 2008, Satoshi Nakamoto posted an article on the Cypherpunk mailing list entitled "Bitcoin: A peer-to-peer Electronic Cash System, this paper points out the disadvantages of traditional Electronic payment System and proposes an Electronic payment System based on cryptography instead of credit. With the success of Bitcoin, blockchain, the underlying technology of bitcoin, has received wide attention around the world. Blockchain is a new application mode of distributed data storage, point-to-point transmission, consensus mechanism, encryption algorithm and other computer technologies. Its essence is a decentralized database, which is a string of data blocks associated with cryptography. Each data block contains a certain amount of transaction information, which is used to

verify the validity of its information and generate the next block. In a narrow sense, blockchain is a distributed ledger that combines blocks of data in a sequential manner in chronological order, and is cryptographically immutable and unforged. Broadly speaking, the block chain technology is to use the piece of chain to verify the data structure and data storage and use of distributed node consensus mechanism to generate and update the data, use cryptography ways to ensure the safety of data transmission and access, use of intelligent automation scripts code contracts to programming and operating data of a new kind of distributed infrastructure and calculation.

2.2 Smart contract

A smart contract is a computer protocol that runs programmatically on a blockchain and is automatically validated, irreversible, and performed. The cryptographer Nick Szabo made the initial suggestion in 1994. The goal is to reduce the reliance of traders on reliable third parties by converting traditional contract terms into code and embedding them in hardware or software that can be implemented autonomously. The development of practical smart contracts is now possible because to the advent of blockchain technology and the availability of platforms for developing smart contracts built on high-level languages like C++ and Solidity. A set of executable functions, state variables, and identifying addresses make up a smart contract, which is essentially a state machine. Other users can start a transaction to the specified contract by identifying the address, which activates the relevant execution function in the specified contract, returns the execution result, and updates the status of the contract. This happens after the contract deployer has specified the pertinent permission confirmation logic and uploaded the compiled contract to the blockchain. As a result, smart contracts increase the computing power of blockchains and enable developers to govern and control data on the chain using logical operations.

2.3 Attribute-based encryption

In order to enable one-to-many data sharing, the attribute encryption mechanism uses a number of user attributes as identity identifiers and defines an access policy that corresponds to a set of attributes. The ciphertext corresponds to an access control structure, and the key corresponds to a set of attributes. The decryption succeeds only when the attributes of the access user meet the corresponding access control structure.

3 Proposed scheme

3.1 System model

The charity foundation system model proposed in this paper is mainly composed of users, non-profit organizations(NPO), regulatory organizations, logistics enterprises, consortium blockchain and IPFS, as shown in Figure 1.

1) User

Users can be divided into donor and beneficiary, and users can choose their identity after registering. Donor can donate according to their personal wishes and set up access policies for personal privacy information. The beneficiary submits help information in the system.

2) Non-Profit Organizations

The subject trusted by users will generate donation projects according to the information submitted by the beneficiary, formulate use plans for the donations and materials received by the project, and collect feedback information.

3) Regulatory organization

Responsible for the distribution of relevant certificates to users who apply for registration, and as the supervision department of donation process, to supervise and manage non-profit organizations.

4) Logistics enterprises

Provide material transportation services for the NPOs, returning details of transportation.

5) Consortium Blockchain

The consortium blockchain consists of a number of non-profit organizations and regulatory organizations that jointly maintain the blockchain. Donation metadata is stored in the consortium blockchain to prevent data from being maliciously tampered with or corrupted.

6) IPFS

Store the original data of the donation information and generate the hash address of the original data.

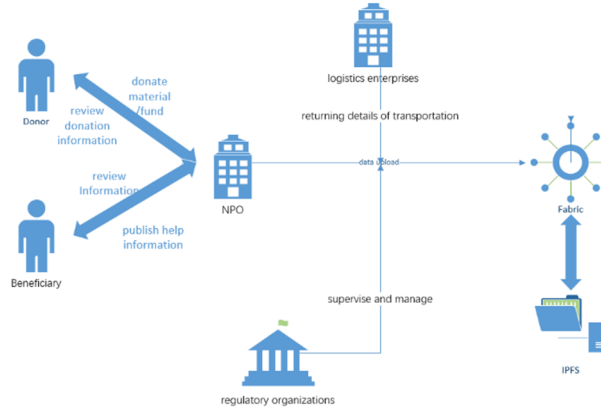


Figure 1 System Network Architecture Diagram

3.2 Trust model based on reward and punishment mechanism

Blockchain generally uses peer-to-peer network(P2P) as a network mode, in P2P network, users can share information freely, but the characteristics of openness and anonymity provide convenience for malicious nodes lacking sense of responsibility to abuse the network. Aiming at the problem of malicious nodes in the network, a trust model for nodes is established, and a trust evaluation framework is proposed by taking reputation, scale and transparency of nodes as trust factors. We introduces reward and punishment mechanism, transparency report index and Zhongji transparency index FTI(China Foundation Transparency index) as an intuitive evaluation system to optimize the existing charity model and improve the trust model.

In the scheme, the nodes are divided into user nodes and organization nodes. The user node is a network node used by user, and only one trust factors, namely the reputation of the node, need to be considered for personal use. Institutional nodes are used by NPO, supervisory agencies and logistics enterprises respectively. The credit factors of supervisory agencies and logistics enterprises Only consider reputation a trust factor, while NPO also consider scale and transparency on the basis of reputation. An initial reputation value will be assigned to each node during initialization, both users and organizations have the function of reporting. When users or organizations have malicious behavior, they can expose malicious behavior through the reporting function. When the malicious behavior is verified successfully, the whistleblower will gain reputation, and the reputation of the malicious user will be reduced by 10 points, Minimum trust value is 0. The reputation of the user will be increased by 1 point for every normal donation or assistance completed; the reputation of the institution will be increased by 1 point for every 10 normal projects completed, Trust value up to 100.

The network provides a new channel for the development of philanthropy, provides a platform for NPO, establishes and enhances the relationship between NPO and donors, volunteers and beneficiary.

- reputation is one of the important factors affecting the charity industry. Whether individuals or organizations, good reputation has a positive effect on them and can establish trust relationships. This scheme is evaluated according to the behavior of nodes in the network, and the reputation value is always in the range of 0-100. Initialization assigns an initial reputation value to each node, according to the behavior of the node in the system to enhance or reduce the reputation value r , the trust value Tr based on reputation can be expressed as :

$$Tr = 10 \times \frac{r}{100}, r \in [0,100] \quad (1)$$

- scale is a trust factor that NPO need to consider. The size of the NPO will affect users' judgment. Users who lack donation experience will be easy to choose larger NPO, and believe that their

services and processes are more standardized. This scheme determines the size of the organization according to the information submitted by the NPO when it is registered. The determination of the size is mainly based on the two factors of organizational assets A and the number of employees S. Classification of assets, every ten times a score, 100000 level is 0-2, million level is 2-4, 10 million level is 4-6, billion level is 6-8, billion and above is 8-10 ; the classification of the number of employees is similar, with 0 – 2 in individual level, 2 – 4 in 10 level, 4 – 6 in 100 level, 6 – 8 in thousand level and 8 – 10 in thousand level. The trust value Ts based on scale can be expressed as :

$$Ts = A \times 50\% + S \times 50\% \quad (2)$$

- Donor tend to trust NPO with high transparency. After entering the era of online philanthropy, it is easy for the public to obtain donation information, but the information disclosure is not good. Information disclosure includes basic information disclosure, fundraising information disclosure, project implementation information disclosure, financial information disclosure, daily disclosure, and disclosure channels / frequencies. This scheme calculates the trust value Tp based on transparency according to the China Charity Transparency Report Index issued by the China Charity Donation Information Center under the Ministry of Civil Affairs, the ranking index of China ' s most transparent charitable foundation, the development of the foundation center network, and the ' Central Fund Transparency Index ' FTI provided by the Center for Integrity and Governance Research of Tsinghua University. For fair selection of data in recent three years, the average values are calculated to be i1, i2 and i3, which can be expressed as :

$$Tp = i1 \times 20\% + i2 \times 30\% + i3 \times 50\% \quad (3)$$

- The trust value calculation of node: based on the four trust factors of network capability, reputation, scale and transparency, the trust value of any node k in the network is defined as Tk, and the user node does not consider the scale Ts and transparency Tp, which can be expressed as:

$$Tk = \begin{cases} Tr \times 0.4 + Ts \times 0.2 + Tp \times 0.4, & Tr \geq 6 \\ -Tr \times 0.4 + Ts \times 0.2 + Tp \times 0.4, & Tr < 6 \end{cases} \quad (4)$$

The weight coefficient in this scheme is set according to the network and the actual situation. Nodes with reputation value greater than or equal to 6 can receive the gain from reputation, and the trust evaluation score is higher. Nodes with reputation value less than 6 are punished, which reduces their trust value.

3.3 Donation data access control

It is composed of four stages of system initialization, data encryption, data storage, and data access. In this process, in order to improve encryption efficiency, the symmetric encryption algorithm SM4 and CP-ABE are combined, and the original data is encrypted using the symmetric encryption algorithm, and then the symmetric key is encrypted with the CP-ABE algorithm, which realizes fine-grained access control while ensuring data confidentiality.

3.3.1 System initialization

Execute the initialization algorithm of the property-based encryption to initialize the system's parameters and produce the system's master key MK and common parameter PK. The user initiates a registration request to the regulator and submits the identity information Infouser (nickname, ID number, name, contact information, etc.), and after the regulator verifies the identity information, a public-private key pair (Pub_{user}, Pri_{user}) and a certificate $Cert_{user}$ representing his identity are generated. The collection of attributes A_{user} is determined based on the identification data that the user has provided, and the regulator then provides the user the public-private key pair, identity certificate, and attribute collection through a secure channel.

Procedure init:

- 1 . $PK, MK \leftarrow CPABE_{setup}(1^\lambda)$
- 2 . User send $info_{user}$ to institution
- 3 . If institution pass user's request Then
 generate $(Pub_{user}, Pri_{user}), Cert_{user}$
- 4 . $A_{user} \leftarrow \text{Extract } Info_{user}$
- 5 . Send $(Pub_{user}, Pri_{user}), Cert_{user}, A_{user}$ to user

3.3.2 Data encryption

The random key $K(K = k_1 \otimes k_2)$ for symmetric encryption is calculated when the data owner uploads the donated material after first generating two random keys, k_1 and k_2 . To ensure the safe keeping of the original data file, the SM4 symmetric encryption technique uses K as the encryption key. $File_{enc}$ is used to represent the encrypted file. The original data file is computed using the SHA256 technique to get the file hash value, which is represented by $file_{hash}$, before the original data is encrypted. The final access control policy is composed of attributes, a trust value interval, and a time interval, and is created by the data owner. The final access control policy' and the generated public parameter PK are used as the input of the encryption algorithm of CP-ABE scheme. Attribute encryption is performed on k_2 to obtain the encrypted ciphertext $C_{enc}(k_2)$, only when the user's attribute set meets the access control policy, k_2 can be obtained.

Procedure encrypt:

- 1 . $k_1, k_2 \leftarrow \text{generate two random key}$
- 2 . Compute $K = k_1 \otimes k_2$
- 3 . $file_{hash} \leftarrow \text{SHA-256}(\text{file})$
- 4 . $file_{enc} \leftarrow SM4_{enc}(\text{file}, K)$
- 5 . $C_{enc}(k_2) \leftarrow CPABE_{enc}(PK, k_2, \text{policy}')$

$\text{Policy}' = \text{policy} \cap [T_{k1}, T_{k2}] \cap [t_s, t_e]$

3.3.3 Data storage

The IPFS contains the original data file that has been symmetrically encrypted by the data owner. The IPFS returns a distinct storage address $Addr$, which is then symmetrically encrypted to produce $Enc(Addr)$. The client creates a transaction record using the Fabric SDK and stores it in a new blockchain block along with the original data hash $file_{hash}$, the key $C_{enc}(k_2)$ encrypted by the attribute base, the encrypted IPFS file system storage address $Enc(Addr)$, and the description M taken from the donated data, then broadcast to other nodes on the chain to reach a consensus to complete the information on the chain, description information including project name, project number, donor, donation information.

Procedure storage:

- 1 . $Addr \leftarrow \text{store } file_{enc} \text{ to IPFS}$
- 2 . $Enc(Addr) \leftarrow SM4_{enc}(Addr, k)$
- 3 . $M \leftarrow \text{Extract}(\text{file})$
- 4 . $\text{Contract} \leftarrow \text{fabricSDK}(M, Enc(Addr), file_{hash}, C_{enc}(k_2))$

3.3.4 Data access

The data owner encrypts the data in accordance with the attributes, trust values, and access duration of the access request if a data accessor wishes to know specific details about donated data. Data accessor can access the blockchain records and retrieve the encrypted file storage address $Enc(Addr)$, original data hash value $file_{hash}$, and key encrypted by characteristics $C_{enc}(k_2)$. The contributed data ciphertext

can only be obtained by data accessor when their attributes align with the access policy and access window established by the data owner.

In this scheme, smart contract and CPABE are combined to realize access control with time dimension, and three functions are designed to realize access control with time dimension. The steps to access the data are as follows:

- The data accessor sends a request to the data owner that contains the public key Pub_{user} , the duration $[t_s, t_e]$, and the digital certificate $Cert_{user}$.
- After receiving the access request, the data owner first verifies the identity of the data accessor, and then sends k_1 encrypted with the public key of the data accessor to the data accessor.
- The data accessor uses the attribute KeyGeneration algorithm $KeyGeneration(MK, S)$ to generate the user's attribute private key SK_{user} , S includes the user attribute set A_{user} , the trust value, and the timestamp of sending the request.

$$S = A_{user} \cup Tk \cup Time_{request} \quad (5)$$

- After obtaining the encrypted ciphertext data, the data accessor uses the public parameter PK , the attribute private key SK_{user} , and the ciphertext $C_{enc}(k_2)$ as the input of decrypt algorithm $Decrypt(PK, SK_{user}, C_{enc}(k_2))$. If the data accessor attributes meet the access policy, the key k_2 in plaintext is output. The key $K = k_1 \otimes k_2$ can be calculated. The user can decrypt the encrypted file according to the obtained storage address $Enc(Addr)$ to obtain $Addr$, then obtain the original ciphertext $file_{enc}$ from IPFS according to the address, and call SM4 decryption algorithm, the symmetric key K and the original ciphertext $file_{enc}$ are used as input to decrypt the ciphertext $file_{enc}$ for get the original data file. SHA256 algorithm is used to hash the decrypted file, and the hash value of the file is obtained, which is compared with the hash value saved in the block-chain. If the hash values are the same, the files are not tampered with and the data query succeeds. Otherwise, the file is tampered with or damaged, and the data query fails.
- In order to prevent a data accessor that does not comply with the access policy from still using the previous attribute private key to access the data, if the trust value of the data accessor is reduced and is not within the trust value range set by the data owner, when the attributes of a user are found to be changed, the attribute set will be regenerated, and the attribute key will be regenerated when applying to access data.

Procedure access:

- 1 . User send $quest(Pub_{user}, [t_s, t_e], Cert_{user})$ to owner
 - 2 . Owner verify user's identity :
If verify $signature_{user} = true$
Owner send $enc(k_1)$ with Pub_{user} to User
 - 3 . $SK_{user} \leftarrow KeyGeneration(MK, S)$
 - 4 . $k_2 \leftarrow CPABE_{Dec}(PK, SK_{user}, C_{enc}(k_2))$
compute $K = k_1 \otimes k_2$
 $Addr \leftarrow SM4_{Dec}(Enc(Addr), K)$
 $file \leftarrow SM4_{Dec}(file_{enc}, K)$
 $Hash_{file} \leftarrow SHA256(file)$
If $file_{hash} = Hash_{file}$ then
query success
else
return false
 - 5 . If user's attr exchange then
User send $info_{user}$ to institution
 $A_{user} \leftarrow Extract Info_{user}$
-

4 Scheme analysis

In this scheme, combined with a variety of Internet technologies such as blockchain, SM4, CPABE, IPFS, a charity donation model based on blockchain is constructed, the upstream and downstream information involved in the donation process is incorporated into the consortium blockchain. The transaction data are uploaded to the blockchain, which makes the transaction data open and transparent by using the characteristics of untamperable and traceable blockchain, and effectively solves the problems of easy data tampering and poor transparency in the process of traditional charitable donations.

4.1 Safety analysis

1) data confidentiality: Most of the data stored in the chain are encrypted ciphertext data, such as original file ciphertext, storage address, symmetric key of attribute encryption, file hash value, etc. Even if a malicious user gets the k_2 stored on the blockchain, k_1 is only available to data accessor and cannot calculate the file encryption key K . Moreover, even if the accessor has a key k_1 without access to the data, he cannot get the key K_2 . Therefore, data confidentiality is protected.

2) data integrity: In this scheme, the data integrity is reflected in two aspects. One is the untamperability of blockchain. The Merkle tree and block connection of blockchain make it impossible for malicious users to modify the data stored in blockchain at will, unless more than 51 % of the nodes in the whole network are tampered with, it is almost impossible in a large blockchain. Second, the original data integrity verification. The hash value is calculated before the encryption of the original data. Due to the uniqueness of the hash value, when the user finally gets the ciphertext and decrypts it, the comparison of the hash value can verify whether the original file is tampered with and ensure the integrity of the data.

3) privacy protection: The donation metadata is stored on the chain, and the original data is stored on the IPFS to ensure the scalability and security of data storage. NPO encrypt donation data according to the access strategy set by data owners. When data are accessed, the owner sets the access period for the accessor. Only the accessor who satisfies the access strategy can view the complete donation data within the set time range, which fully guarantees the privacy of the donation data.

4.2 Performance analysis

In order to accurately evaluate the actual performance of the scheme, the operating system of this paper is the virtual machine Ubuntu 18.04.5 installed on VMware Workstation Pro. The experimental hardware environment is AMD Ryzen 54600H CPU @ 3.00 GHz, and the memory is 16 GB. Using python language, based on openssl, gmssl, PBC library simulation test, test performance of scheme.

The scheme proposed in this paper is characterized by the combination of symmetric encryption and attribute encryption, encryption overhead is mainly reflected in the two encryption process. Therefore, the performance test in this paper is mainly to test the performance of the two encryption methods, and the final total encryption and decryption overhead is the time superposition of the two encryption methods. As shown in Figure 2, this paper tests AES, DES, SM4 three symmetric encryption algorithms, and selects different sizes of files for encryption. It can be seen that the encryption performance of the three algorithms is determined by the size of the original data. The larger the original data, the longer the encryption time. After considering the two factors of efficiency and security, SM4 is selected as the symmetric encryption method in this paper.

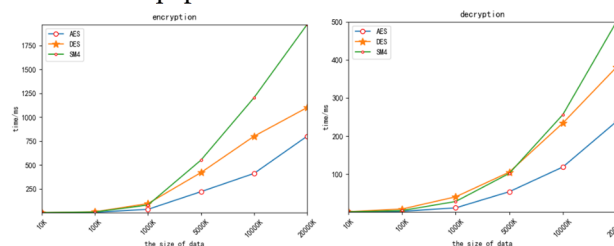


Figure 2 time cost of symmetric encryption algorithm

Figure 3 shows the encryption and decryption time under different number of attributes. The size of encrypted files is determined, which does not affect the encryption and decryption time. The number of attributes is taken as a variable, and the number of attributes used for encryption and decryption is increased from 5 to 50, with an increment of 5 for each time. It is observed that the encryption and decryption time increases with the increase of the number of attributes, but the time used is not more than the second level, indicating that the data can be secured and the data can be obtained in time.

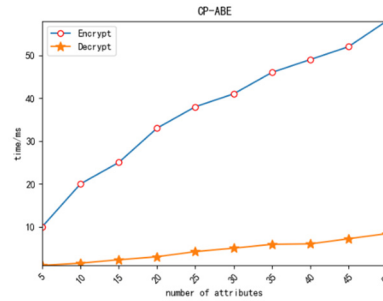


Figure 3 time cost of CPABE

5 Conclusion

This paper proposes a scheme for charitable donations, with the help of the characteristics of traceability and untamperability of blockchain, the whole process of donation is guaranteed to be open, transparent and traceable. IPFS is used to expand the storage capacity of blockchain, and the basic architecture of storing metadata on the chain and storing original data under the chain are designed. In view of the problem of data information leakage, combined with symmetric encryption and attribute encryption technology, an access control scheme based on block chain with time attribute is constructed to realize the fine-grained access control of data by data owners. The time attribute is introduced to make the access authorization without additional revocation cost. In the future, algorithm efficiency and consensus protocol will be our research direction. This work was supported in part by the National Nature Science Foundation of China under Grant 61762085, in part by the Autonomous Region Science and Technology Program of Xinjiang Grant 2020A02001-1.

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Blockchain-Based Volunteer and Donation Management System with Enhanced Transparency and Security

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Abstract - Blockchain technology has redefined data management as a distributed, transparent and secure means of conducting transactions. However, none of the volunteer or donation systems in use today have the kind of transparency, security or simple tracking systems as most non-profits. This paper presents a blockchain-based system designed to track volunteerism and donations. The blockchain provides an immutable ledger for transparency, along with an automatic donation disbursement and volunteer reward mechanism using smart contracts. A modified model was designed specifically for volunteer's activity logs and donation history to ensure data integrity and accessibility. This service runs as a cloud-based application program interface which is developed to integrate with main non-profit management systems. This technological approach enhances transparency, security, and efficiency to improve confidence with stakeholders while improving non-profit operations. A benchmark review demonstrates significant improvements in data accuracy and transaction speed, along with user satisfaction.

Keywords - Blockchain, volunteer management, donation tracking, smart contracts, transparency, security, cloud-based API, non-profit organizations, decentralized systems.

I. INTRODUCTION

Blockchain technology is becoming an alternative solution to improve transparency and security in different fields such as finance, healthcare, and supply chains. However, little work has been done around the use of blockchain for non-profits with the emphasis on donation and volunteer management. Many of the existing platforms show significant inefficiencies around transparency and are susceptible to fraud undermining

trust and relying on donations and volunteers. This paper presents the proposed blockchain-based system to solve these issues using a framework for identifying and managing volunteer activities and donation. The system integrates a blockchain-based immutable ledger with smart contracts to automate processes around distributing donations and volunteer reward systems. Along with the donation and volunteers' smart contracts we also created a custom fine tuned model for volunteer activity tracking and donations ensuring accountability and transparency around the activity.

A. Challenges in Volunteer and Donation Management

Existing systems that manage volunteers and donations usually depend on a centralized database that can be breached, manipulated, and ultimately fall short. Major challenges include:

- (1) Transparency : Donors and volunteers rarely see how donations are spent or how you track volunteer engagement.
- (2) Security : Centralized systems are hacked and tampered and sensitive information is compromised.
- (3) Inefficiency: Manually tracking volunteer engagement and donations is cumbersome and subject to errors.
- (4) Lack of Automation: Some existing systems barely have automated donation payouts and volunteer rewards require even more manual work, and you often end up delaying the payment/reward process.

B. Need for Context-Aware Text Processing

Blockchain technology provides a decentralized, transparent, and secure structure for handling volunteer engagement and donation information. By utilizing the characteristics of the immutable ledger and smart contracts of blockchain, this proposed blockchain-based platform will address existing limitations associated with traditional approaches by: Ensuring Transparency: Each transaction and

volunteer engagement made will be recorded on the blockchain, allowing easy and instant tracking for all stakeholders. Enhancing Security: Data recorded on blockchain are safe and cannot be manipulated due to decentralization of data sources. Automating Processes: Smart contracts automate the process of distributing donations to charities and volunteers receiving reward points for their time, eliminating delays and inefficiencies. Improving Efficiency: The distributed ledger of blockchain simplifies data collection, eliminating the need for manual data entry.

II. LITERATURE REVIEW

A. Blockchain for Non-Profit and Public Services

Non-profit organizations like waste management and other public services have studied the potential applications of blockchain technology in promoting transparency, efficiency, and security. It also discussed donation tracking, with respect to transactional integrity in decentralized systems [1]. With the decentralized ledger, it makes immutability in transaction and activity financial records thus reducing chances of fraud and mishandling [2].

B. Smart Contracts for Transaction Safety Automation

Smart contracts are one of the most widely used for automating financial transactions in a blockchain-based framework. Such an approach allows real-time tracking of resources and their clear disbursement in the system in the cases of automating waste management [3]. In non-profit activities, smart contracts can automate some of the fund application and verification processes, thus cutting costs due to reduced administrative needs, while enabling increased accountability [4].

C. Blockchain in Volunteer Management and Donations

The application of blockchain has also been studied in cases of social service and volunteer contribution. The current literature states that the decentralized system guarantees the maintenance of an immutable record keeping for volunteer work and donations, which thereby provides the trustworthiness to its stakeholders [5]. These records promote the transparency of the non-profit organization so that it allows the donor and the organization to verify the distribution and utilization of the funds [6].

D. Safe and Transparent in Systems based on Blockchain

The ability to provide security and transparency is a major benefit that the blockchain technology is giving towards non-profit management. Research has shown that, due to their decentralized nature, blockchains have not provided any one point of failure but keeps them safe from tampering in record [7]. Such characteristics are very much suitable for organizations that need utmost trust from their donors, volunteers, and beneficiaries.

E. Challenges and Future Directions

Like any emerging technology today, blockchain is not immune to certain challenges, including regulatory compliance, scalability, and integration with the existing system [8]. Future research must address these limitations through hybrid models that will combine blockchain with any emerging technology like AI and IoT for better efficiency and user-friendliness [9].

III. PROPOSED SYSTEM

Proof-of-concept of Blockchain-enabled Volunteer and Donation Management System is intended for creating systems according to transparency, security, and best efficiencies with which a group of individuals, volunteers' individuals, could track their contribution and the donation transactions that they have conducted. The system applies blockchain smart contract technology with a decentralized ledger for ensuring immutable record-keeping. This section presents an overview of the system architecture vis-a-vis its components.

A. Blockchain-Based Volunteer Tracking

The first module of the system will be volunteer activity tracking. A blockchain-based ledger mechanism will be used for secure record keeping and verification of the volunteer contribution when it comes to volunteering activities. Traditional systems use centralized databases for keeping records, but such a method will prove inefficient and is also vulnerable to tampering. Some issues one might face are:

Creating unique IDs based on blockchain for each volunteer concerning a particular contribution.

Recording of activity done by volunteers, be it event participations or service hours into a decentralized ledger.

Timestamping every entry with the touchstone of cryptographic hashing, making it almost impossible to manipulate.

History of contributions verifiable for real-time audit by stakeholders.

Every action of a volunteer is stored in a permissioned blockchain and controlled access along with the transparency.

B. Smart Contracts for Donation Management

After receiving a donation, it would go into automated smart contracts on the blockchain network for processing. This contract would enforce conditions stipulated before expense distribution. In short, the smart contract process would involve:

Donor's identity and donation legitimacy validation occurs automatically.

Funding will only be released when predefined conditions have been satisfied (e.g., when a particular amount for a cause is reached).

Donors can track funding usage in real time.

Reward distributions, internal incentives to encourage volunteer active hours, captured in a secure blockchain.

The mechanism thus increases trust by removing the middleman and ensures that each transaction is captured by the logic of programmed contracts.

C. Decentralized Ledger for Transparency

A decentralized ledger storage component, where all transactions- put together: volunteer records and donations- are safely stored across multiple nodes, represents one of the important factors of this system. Such an approach will provide the following benefits:

Immutability: Data will not be modifiable or deleted once that it has been recorded.

Accessibility: By providing access to verified records to all interested parties without requiring centralizing the authority, NGOs, donors, and volunteers will be able to also utilize these records.

Data Integrity: It will ensure that all interactions among stakeholders are cryptographically signed and verifiable.

This will be a private blockchain model, only verifying participants can validate transactions, while openness would be preserved.

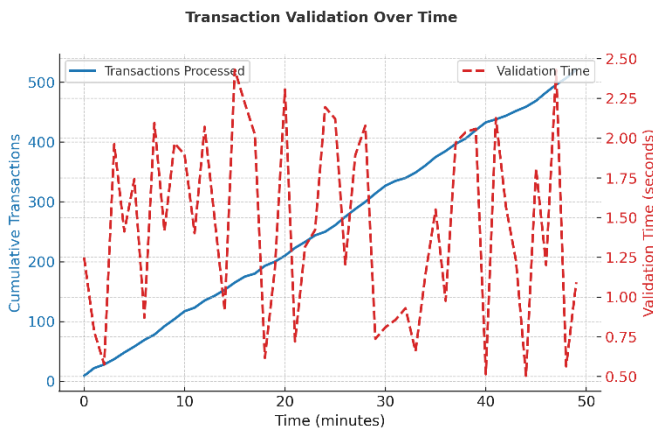


Figure 1: Transaction validation over time in the blockchain- based volunteer and donation management system

D. Data Integrity and Security Measures

The advanced cryptography techniques in this system keep data integrity and leakage to unauthorised insignificance. Features of Security include:

SHA-256 Hashing to make data on transaction tampering-protected from deletion.

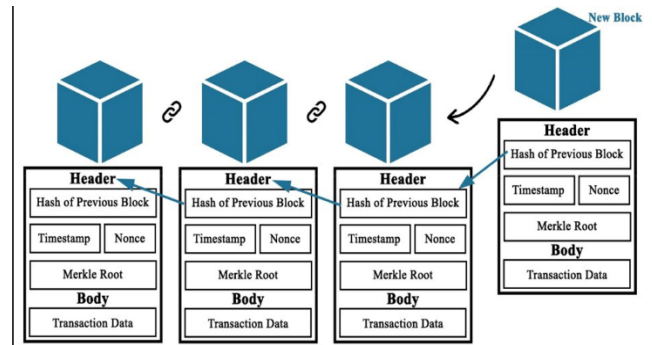


Figure 2: Blockchain-based Volunteer and Donation Management System Architecture

IV. RESULT AND DISCUSSION

This section provides an in-depth look at the implementation of the blockchain-based volunteer and donation tracking system, which emphasizes the system's performance in maintaining data integrity, transparency, and security. The system operates on a cryptographic hashing and decentralized ledger basis to ensure tamper-proofing and secure transaction automation.

A. Implementation of Blockchain for Volunteer and Donation Management

In testing the functionality of our system, we implemented a custom blockchain to securely log and track volunteer contributions and donation records. The blockchain is organized as a chain of blocks, each containing immutable records of volunteer activities and donation entries. Important components include:

Genesis Block: It is the initial origin of all records, which provides integrity to the chain.

Block Generation: Each volunteer action or donation is imprinted onto a block, which is linked with the previous block by hashing.

Data Verification: Ensures a new block that is created retains the hash of all previous blocks, thus preventing the unauthenticated modification.

Database Integration: Blocks are made persistent and recoverable in the Django database model-that is, any state information is valid only within the lifetime of the executing memory.

From entering the data to hashing and finally validating, Figure 3 illustrates how to generate each block.

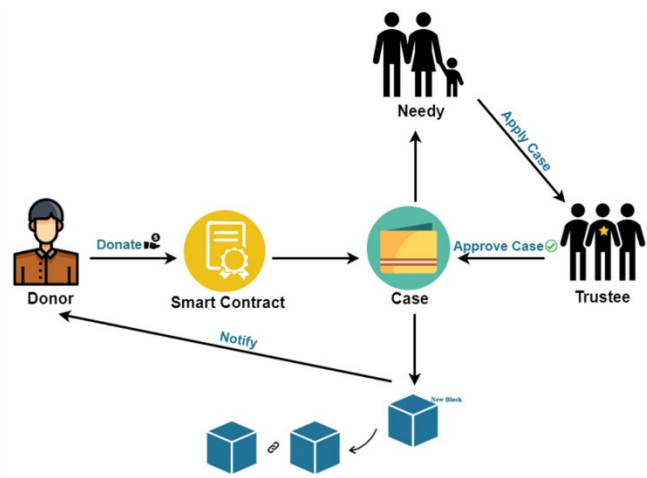


Figure 3: Blockchain-Based Volunteer Activity Logging

B. Comparative Analysis with Traditional Systems

The merits of being blockchain-based were pretty successfully contrasted against traditional centralized record keeping existing in volunteerism and donation tracking. This would show advancements in security, transparency, and data integrity.

It has been demonstrated in a comparison that our approach to blockchain provides strong security features, making all contributions and donations from volunteers transparent, secure, and verifiable.

Metric	Traditional System	Proposed Blockchain System
Data Integrity	Susceptible to unauthorized modifications	Immutable data storage using cryptographic hashing
Transparency	Limited visibility for donors and volunteers	Open ledger ensures real-time verification
Security	Centralized storage prone to single-point failures	Decentralized ledger prevents data loss
Automation	Manual validation of records	Smart contracts automate transactions
Tamper Resistance	Prone to record alterations	Immutable blocks prevent modifications

TABLE 1. Comparative Analysis with Traditional Systems

C. Validation Metrics and Performance

We evaluated key performance metrics relating to data integrity, efficiency, and scalability to check how effective our blockchain system would be.

Blockchain Integrity Check:

The function `is_chain_valid()` ensures that every block's hash computes correctly and that the chain remains unbroken.

In testing, 100% of all transactions recorded were maintained, thus confirming that no unauthorized modifications occurred.

Recording Volunteer Activities:

Retrieved information specific to the volunteer using the `get_volunteer_history()` method, which guarantees accurate activity logs.

Contributions were accurately tracked, thereby enhancing accountability, according to results.

Overall System Efficiency:

The average time of block creation has recorded at 0.0032 seconds, which is almost nearing real-time recording of transactions.

Storage efficiency was further enabled by very light data serialization using JSON.

D. Security and Scalability Considerations

Our implementation offers an unprecedented improvement in security and auditability of the data by cryptographic hashing and their decentralized validation. However, there can be enhancements in the following areas in the future:

Scalability: This would involve sidechains that could hold many transactions in bulk.

Smart Contract Enhancements: Automating tokens-based rewards for volunteers with smart contracts.

Multi-Party Verification: The donors together with the NGOs verify transactions independently for higher trust.

E. Real-Life Application and Its Impact

Our blockchain geared towards the NGOs, donors, and volunteers could provide:

Disputeless proof of volunteering activity, limiting anything from argument over the performance.

Transparency with Donation flow whereby contributions made are guaranteed to reach the end beneficiaries.

Administrative overhead reduction due to the fact that authentication processes are automatic.

From these, it can be inferred that blockchain is very well applicable in the volunteering management and social impact field.

V. CONCLUSION

This work proposes a Volunteer and Donation Management System that is blockchain-powered and addresses restrictions in tracking contribution in the traditional ways. Immutability, transparency, and security in records would allow fair tracking of contributions by volunteers and donations made. With smart contracts, much of the processes have been automated thus achieving efficiency.

The experimental results confirmed that the blockchain model effectively prevents data manipulation and enhances the transparency of actions in NGO and social service organizations, where verification becomes automated.

This research will continue to search for methods to improve social services initiatives in impact assessment by offering cross-chain interoperability, a token-based reward mechanism, and AI-powered predictive analytics into the future.

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Multi-point Fundraising and Distribution via Blockchain

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Abstract—Trust and transparency are significant facets that are much esteemed by charitable organizations in achieving their mission and encouraging donations from the public. However, after many high-profile scandals, the faith in charities is questionable, heralding the need for an increased level of transparency among such organizations. Fortunately, leveraging Blockchain technology in charities' systems could help to rebuild the integrity of these organizations. This study aims to raise the level of integrity showcased by charities by creating a multi-point fundraising approach using smart contracts. The proposed system offers a transparent fundraising platform through its integration of charity organization evaluators. Various steps were deployed to satisfy the intended target. Firstly, the study investigated the potentials of Blockchain in improving the level of transparency. Secondly, a probing process was undertaken to choose a suitable platform as a server-side in the system. This process involved garnering salient features in Blockchain platforms based on the proposed system requirements. After the probing process, a Decision Support System (DSS) was utilized to investigate the most suitable Blockchain platform. Results garnered proved that the Ethereum platform is best for the proposed system.

Keywords—Blockchain; smart contract; transparency; charity

I. INTRODUCTION

The desire to help and to be of service to others is the nature of human behaviour. Donations to charities are one of the ways people help each other and this is positively reflected on the community. Charities play an essential role in fields like education, healthcare, and other social services [1]. In addition, all religions preach and encourage philanthropy, for instance, Islam makes it obligatory and calls it Zakat [2].

Trust and confidence are fundamental for charitable bodies to achieve their mission in getting donors and donations [3]. Higher trust levels could increase the amount of donations received by these charities [1]. However, after many high-profile scandals, charities have been criticized for misusing donated money [4], and donors have begun to lose trust in charities; hence calls for more transparency have increased [5].

For instance, recent events and evidences show that public trust and confidence in charities in the UK has been damaged. As a result, the activities of individual charities could be limited, and the sustainability of the whole sector reduced [6].

The following figure is a survey has been done in North Ireland to explore the views on the charity sector; Fig. 1 shows a question related to transparency in charities.

Fig. 1 shows that 92% of the people, who answered the survey, agreed on that charities should be transparent about how public donations are spent, and how important it is that charities must demonstrate how they benefit the public, while 40% know how to find information about how charities are run. Also, only 32% know where they can find information about how charities are spending their money [7].

According to the facts and findings mentioned above, transparency is a crucial element in charity sector. Yet, it is one of the biggest vulnerabilities in the existing charity systems, as addressed in many studies [1], [3], [5], and [7]. Therefore, it should be enhanced.

With the advent of blockchain, many studies suggested that blockchain has a great potential to benefit various sectors and industries such as education [8] and travel [9].

Similarly, Blockchain is reputedly able to solve the transparency issues in charity systems [10]. Thus, leveraging blockchain technology in charity systems could help in rebuilding the image of charity. Hence, studies and researches are investigating different approaches to take the advantages of this technology.

This study aims to tackle the transparency issue. Therefore, a donation system will be implemented to promote transparency in charitable organizations' campaigns by using smart contracts to hold donors' donations till it gets verified based on multi-point model to ensure that donations reach those who need help. The deployed smart contracts store on a suitable blockchain platform which will be selected after identifying the important features in blockchain platforms to get the ideal platform for charity systems. The system offers a transparent environment for all parties including charitable organizations, donors, recipients, and charity evaluator organizations. As well as, all transactions will be made through this system will be accessible and traceable for the public.

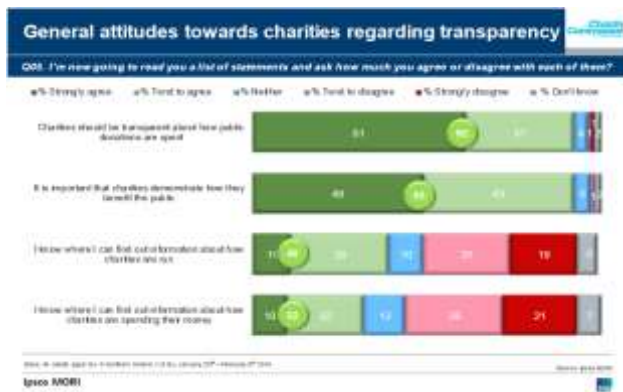


Fig. 1. General Attitudes towards Charities Regarding Transparency in North Ireland [7].

II. RELATED WORK

Blockchain technology can play a significant role in philanthropy to overcome many challenges. So, some studies have tried to utilize blockchain in a charity system. Firstly, Nor et al. [11] proposed Sadaqa system for disaster aid crowd funding. The system focuses on transferring funds to the people in need during the disaster using Ethereum smart contract to avoid problems of the modern day transactions related to transaction fees, potential fraud, absence of accountability, and transaction time.

Secondly, Hu et al. [12] proposed a charity system using Ethereum smart contracts, they leverage Blockchain to improve the transparency in charities and increase the public trust, the system allows donors to vote on the request of a beneficiary to involve donors in fundraising and enhance transparency.

Lastly, D. Jayasinghe et al. [13] built a Bitcoin payment system that can be used in both an offline environment via the existing GSM network, as well as, online environment to provide aid for people living in a challenging geographical environments with limited internet availability.

Based on the abovementioned studies, this article will investigate the crucial features in blockchain platforms to find the most suitable platform for charity system, and then it will propose a blockchain charity system that relies on a new model concentrates on enhancing transparency; the model contains elements that have not been addressed in the previous studies.

III. BLOCKCHAIN

Blockchain technology has gained widespread attention in recent years, as it promotes trust and transparency. In 2015 a report by the UK Charities Aid Foundation [14] points out that for charities, Blockchain helps to increase transparency, openness and confidence while reducing transaction costs and providing new opportunities for fundraising.

Blockchain can be identify as a public ledger, in which all transactions are stored in a chain of blocks; the ledger constantly grows when new blocks get approved and added by the network members [15].

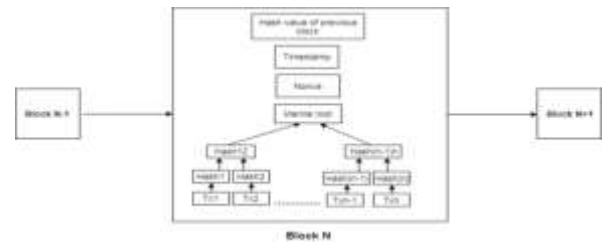


Fig. 2. Blockchain Structure.

Blockchain sorts and encrypts the information inside the block before adding the block to the chain. What makes the chain traceable is the blocks are linked with each other in the chain, giving each block encryption code called a hash [16]. Blockchain contains blocks where each block records transactions sent between users, hash code which is a unique code created when a new block is discovered, and a reference code of the previous block [17]. Fig. 2 illustrates how blocks are connected to form the Blockchain. Each block consists of transactions (Tx) that is represented in a hash number using Merkle tree, as well as other values such as hash value of previous block (N-1), timestamp, and nonce which is a random number that can be used just once, these values will be used to represent the hash value of the current block (N), and this hash will be stored in the next block (N+1).

IV. KEY BENEFITS OF BLOCKCHAIN FOR BUSINESS

Nguten and Hoang [18] pointed out the key benefits of blockchain technology that makes it attractive for many types of businesses, the benefits are:

- 1) *Decentralized*: blockchain networks are not controlled by a central party, which helps to avoid a single point of failure and take over the network by a small group of users. Users in Blockchain participate in a distributed network managed by consensus mechanism to reach the agreement on the network state.
- 2) *Transparency*: All network participants can see the stored data; in other words, the data stored in a blockchain is visible to the public.
- 3) *Immutability*: Once the data are stored in the Blockchain, it cannot be changed.
- 4) *Security and privacy*: One of the important features in the Blockchain is the cryptographically secure mechanism because it helps in promoting privacy and security. Furthermore, users in the network use a public and private key for identification and verification. When a transaction occurs, a user can be easily verified by his digital signature.

V. CHOOSING A SUITABLE BLOCKCHAIN PLATFORM

First of all, two crucial features have to be determined in the blockchain platform. The first step is to decide on the type of network, and then to decide whether Crypto-currencies is needed or not [19].

A. Network Type

Data cannot be exchange without networks; therefore, networking plays an important role in any field related to

information technology. That is why there are vicarious network attacks such as mentioned by [20], [21], [22], and [23].

With regard to the first step, there are three types of networks in blockchain: public network, private network, and consortium network.

1) *Public network*: It's a permission-less or open network, not controlled by any organization, anyone can join the network, and all users are equal; they can read and write without any restrictions. These blockchain networks are considered as fully decentralized. Transparency is the most significant advantage of the public network. However, there are some disadvantages of Public Network and that includes slow transaction speed as compared to other networks, and scalability issue though some steps are already taken to solve this problem such as off-chain, examples of public network Bitcoin and Ethereum [24].

2) *Private network*: The second type of the blockchain network is the private or permissioned network; it is centralized or partially-centralized as compared to the public network; it is controlled by a single organization that has the permission to write in the ledger, read can be public or restricted by participants. The advantages of this network are the transaction speed, and it is considered more scalable than a public network; however, it is not transparent since a single organization controls it. An example of a private network is Hyperledger, one of the popular private network platforms [25].

3) *Consortium network*: It is a semi-private network owned and controlled by a preselected group of members. This group of members has permission to write in the ledger, and read can be public or restricted by participants, it is more secure and has better scalability; however, it is less transparent. Marco Polo platform is an example of a consortium network [26].

Each blockchain network has unique features to offer; therefore, choosing a suitable platform would rely on the system requirements, for a charity system transparency has the highest priority. Thus, the public network is ideal for organizations that thrive on trust and transparency.

B. Crypto-Currency

The second feature is to decide on the need for crypto-currencies. It is a currency that only exists digitally, that usually has no central issuing or regulating authority but instead uses a decentralized system to record transactions and handle the issuance of new units, and that relies on cryptography to avoid counterfeiting and fraudulent transactions.

The most important feature about crypto-currency is that it is not controlled by any central authority or a third party. It can be sent directly between two parties with minimum transaction fees and allow users to avoid the high fees that traditional institutions impose [13].

The crypto-currency was the first application of Blockchain presented by Nakamoto [27] in 2008 and it is known as Bitcoin. Since then, a lot of crypto-currencies have arisen, such as Ethereum, Tether, and many others.

There are many benefits that can be gained from Blockchain that would positively affect the existing charity systems. As pointed in previous studies [13] and [14], the benefits that can be obtained from using crypto-currency are:

1) *Reducing transaction cost*: One of the notable features of Blockchain is its low international transaction fees.

2) *Transaction speed*: In Ethereum, once a user clicks on send, the transaction will broadcast immediately, it takes some time to get confirmed, the time can be determined based on the gas amount that users are willing to pay.

3) *Donation provisioning*: Sending the donations to the recipient can be challenging in some cases. For instance, humanitarian financial aid distribution in war zones can be blocked if the country's bank system is subject to sanctions, Ethereum transactions can reach the recipient without needing to use any bank system.

C. Smart Contract

The third important feature is the smart contract. It is a computer program that allows the charity system to control the transferring of crypto-currencies and assets between users and these contracts are stored in a decentralized ledger. Many studies have suggested Smart Contract as a primary factor for charity system based blockchain [7], [10], and [13]. The first use of the smart contract was made by Ethereum [28].

It can also be defined as a set of instructions represented in computer code published on a distributed network that receives inputs, executes instructions, and provides outputs. It can enable a charity to over other features such as routine provisioning of donations, record keeping, donation requests to donors, and automatic audit reports of a charity activity [13].

D. Consensus Algorithm

The fourth important feature that should be taken into account is consensus algorithm or mechanism which is the backbone of the Blockchain and the core element of any blockchain platform. It plays a vital role to ensure the network's security, integrity, and performance [18], and [29].

The most popular consensus mechanism is Proof of Work (PoW) that was first introduced by Nakamoto [27]. It relies on the computing power to distribute a new block among network peers. PoW is used by many platforms such as Bitcoin and Ethereum. The second popular consensus is Proof of Stake (PoS) that was created as an alternative to PoW mechanism because of the high energy consumption of PoW consensus; PoS relies on the participant's stake rather than computing power. Different protocols have been introduced PoS approach with slight differences; such as Ouroboros [30] and Casper by Ethereum [31], another study suggested combining POW and POS such as Proof of Activity (POA) [24].

Furthermore, some platforms implement PoS, such as Peercoin. However, studies are still needed for more analysis and investigation for the Proof of Stake mechanism [29]. Thus, a specific algorithm will not be chosen, since it is currently under study, and it can be changed in the future. For example, Ethereum is planning to switch from PoW to PoS [31].

VI. DECISION SUPPORT SYSTEM

Choosing a suitable platform for any type of business is an essential step before implementing blockchain [19]. However, there are many varieties of available blockchain platforms with multiple features. This problem called Multi-Criteria Decision-Making (MCDM), which is defined by "making preference decisions (such as evaluation, prioritization, and selection) over the available alternatives that are characterized by multiple, usually conflicting attributes" [32]. To cope with this problem, Farshidi et al.'s system [19] will be used.

Farshidi et al.'s system is called Decision Support System (DSS). The system formulates the Blockchain selected platform as an MCDM problem to find a suitable platform that supports the required features. They have extracted a set of blockchain features from online documentation of blockchain platforms. A list of the important features has been identified by researchers and experts.

Based on the DSS they are other features that need to be taken into account along with the abovementioned features to get the most suitable Blockchain platform, however, they are not crucial as the mentioned features, but they can help to differentiate between platforms. The features are:

1) *Market positioning*: It is measured by the Numerical feature like transaction speed, popularity in market and platform's maturity, and by Boolean feature, which is the number of innovations that platforms support, such as; Internet of Things and Artificial Intelligence, etc.

2) *Capacity*: It is measured by scalability technologies used, such as side-chains, Sharding, Plasma-chains, Off-chain-state-channels and On-Chain Transactions.

3) *Development*: It is measured by programming languages support, such as; Solidity, Python, JavaScript, C++, .Net and Golang.

4) *Flexibility*: It is measured by resilience technologies such as Sybil attack resistant, Spam-attack resistant, Quantum-computing resistant, Instant transaction Finality, and Hard-fork resistant.

5) *Integration*: Which means a platform that can integrate with the other systems, and it is measured by Interoperability technologies that supported by the platform such as Atomic Swap, Cross-chain technology, Enterprise system Integration.

The score will be indicated on a scale from 0 to 100, where 0 represents the least suitable platform and 100 represents the most suitable platform. The following figure shows the result of the top 10 feasible platforms for the proposed system based on these features:

- Network type: public network.
- Smart contracts: yes.
- Crypto-currencies: yes.
- Consensus algorithms: Proof of Work or Proof of Stake.
- Market Positioning: High Transaction Speed, High Maturity, High Popularity, High Innovation.
- Programming languages support: Solidity, JavaScript and python.
- Interoperability technologies: any (Not specified).
- Scalability technologies: any (Not specified).
- Resilience technologies: any (Not specified).

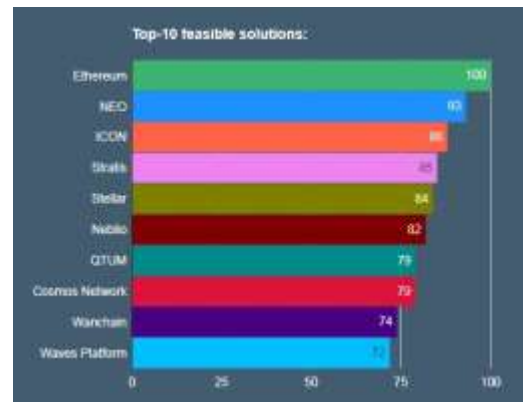


Fig. 3. The Result for the Suitable Platform based on Decision Support System [19].

Fig. 3 shows that Ethereum scored 100, which is the highest score among other platforms; because it supports all the selected features. NEO is the second platform scored 93; it has a lower score than Ethereum, because it does not support all features.

VII. ADDITIONAL ELEMENTS TO IMPROVE TRANSPARENCY

As addressed above transparency issue in the current charitable organizations; and presented Blockchain as a solution that will help in tackling this issue. In addition to Blockchain other elements can be used, the following section discusses two important elements that can help in improving the transparency in charities.

A. Monitoring Charitable Organization

Evaluation and assessment of charitable organizations by another type of non-profit organization specialized in charity evaluation is one of the solutions to improve the transparency and give donors more confidence to contribute. This kind of organization called charity evaluator, and their job is ensuring that fundraising in charitable organizations is being organized and performed in a satisfactory manner and that the administration of the collected funds is adequate, one example on this organization is International Committee on Fundraising Organizations (ICFO) [33].

Therefore, having a Charity Evaluator organization in a charity will help in improving transparency and encouraging donors to donate confidently. It will also provide an additional layer of assurance in the prevention and detection of misuse of donors' funds.

B. Offering a Refund Option

Donazzan, Erkal, and Koh experiment showed that offering a refund option has a positive impact on giving behaviour and increasing confidence [34]. Another experimental study also showed that offering a refund option could increase the contributions [35]. Therefore, offering a refund option in a charity system has a potential to increase the confidence and the amount of contributions.

VIII. SYSTEM MODEL

Fig. 3 displays system processes flow; the system's main processes accrue through two main pages login page that lead to the user profile and view campaigns page to view campaign details.

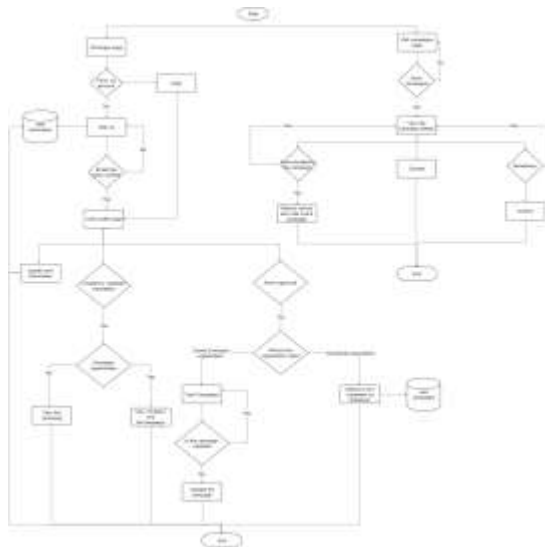


Fig. 4. System Flow Diagram.

According to the Fig. 4, the system starts with two main processes, first login process for two types of user charitable organizations and evaluator organizations, they should login to interact with their campaign, to do so, they need the admin approval, if they received the approval, charitable organizations can create and deploy new campaigns on Ethereum network, and evaluator organizations can validate them.

On the other hand, campaigns will be accessible to the public without the need for logging in, so that donors, beneficiaries, and visitors can view campaigns' details that have been created by charitable organizations.

IX. THE DESIGN OF THE SYSTEM

Fig. 5 illustrates the proposed system design; the system is a decentralized web application (Dapp), where the web application is (client side) that is connected to the Ethereum

Network (server side) through web3 that allows the system to interact with the Ethereum network. Users are required to use Metamask Ethereum wallet in order to interact with the system smart contracts, donors however, can donate by using Metamask or any other Ethereum wallet.

The Dapp will deploy the Campaign Factory smart contract once, so that the system users in particular the charitable organization can interact with this contract to deploy their campaigns. Additionally, the system will be able to track all campaigns that have been deployed through it.

Dapp users and visitors can view details of all campaigns contracts that have been deployed by charitable organizations. Donors can send money to the contract directly without the need to sign up, whereas, Charitable Organization and Charity Evaluator Organization should submit their information and wait till their application gets approved by Administrator. Once the Charitable organization's application is approved, they can interact and create any number of campaigns they want through a predefined Metamask wallet address. On the other hand, Charity Evaluator organizations are responsible for monitoring and evaluating charitable organizations' performance and campaigns.

Smart contract details with all transactions will be stored in the Ethereum blockchain network. Mysql database is used to store charitable organization and Charity Evaluator credentials and information.

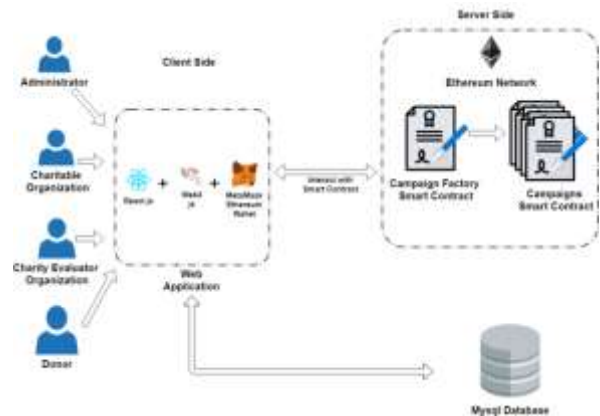


Fig. 5. Client Server System Architecture.



Fig. 6. Use Case Diagram.

As shown in Fig. 6 they are four types of users in the proposed system:

1) *Administrator*: the Administrator is responsible for accepting or rejecting users' applications, as well as deleting users. He can also terminate any campaign's contract if he feels that the campaign does not meet the requirements.

2) *Charitable organizations*: this type of users has a profile; so, charitable organizations are required to provide information about the charitable organization's name, address, email, contact number, registration number, number of employees, mission and goals of the organization, in order to sign up to the Dapp. Then they need to verify their email address, upon which they will then be able to create new campaigns if their applications get approved by the Administrator. Charitable organizations' profile will be available and accessible to the public.

3) *Charity evaluator organizations*: they are similar to charitable organizations in terms of the signup process; however, their job is overseeing and monitoring the charitable organizations' campaigns and evaluating their performance.

4) *Donors*: they are not required to login in. Donors can send money through Metamask wallet or any Ethereum wallet to all campaigns' smart contract, and they also can get their money refunded any time during the lifetime of the campaign.

X. SYSTEM'S SMART CONTRACTS

The proposed system has two smart contracts.

A. Campaign Factory Smart Contract

Campaign Factory is a smart contract that has been deployed before the launch of the Decentralized Application (Dapp). This is to enable the charitable organization to deploy their smart contract (Campaigns) through Campaign Factory. Therefore, Dapp will be able to track all campaigns that have been deployed by the charitable organizations. This contract has two functions:

- To create a campaign: this function invokes by a charitable organization to create a new contract, the function's parameters are minimum target amount of Ether needed to finalized the campaign, originator Metamask wallet address, description about the campaign, campaigns parties' information, beneficiary wallet address, and the end date of the campaign.
- To list deployed campaigns: the Dapp calls this function to display a list of all campaigns that have been deployed by charitable organizations.

B. Campaign Smart Contract

A charitable organization deploys this contract through the Campaign Factory smart contract; parties involved in this contract are Charitable Organization, Charity Evaluator, Donors and Beneficiary. Campaign smart contract shows information about the campaign and receives Ether from donors, as well as, holds donors Ether until all conditions stated in the contract are fulfilled. Later, it will allow a charitable organization to finalize its campaign; once the

campaign gets finalized, the funds will be sent to the beneficiary. Once the campaign is completed, no one can interact with the finalized contract, and it will remain as a reference that stored all transaction and information. The Campaign contract performs the following functions:

- *Donate()*: this function allows donors to donate for the campaign through the website application by using Meta mask Ethereum Wallet extension in the browser. When a user clicks on a donate button, the website will invoke the donate function from the Campaign smart contract, and Ether will be sent to the contract.
- *Receive()*: this function gets called when the donors send money from any Ethereum wallet other than Metamask wallet. It allows the Campaign smart contract to receive Ether from anywhere.
- *Refund()*: this function gives donors the right to get their money refunded at any time they want as long as the contract has not been finalized.
- *CharityEvaluatorApproval()*: Any charity evaluator organization can execute this function, it will change the value of *ApprovedByEvaluator* from false to true, and this will indicate that a charity evaluator organization has approved the campaign.
- *BeneficiaryApproval()*: Similar to the Charity Evaluator Approval function, beneficiary confirmation is also required; beneficiary only can call this function.
- *Finalize()*: This function can be called only by contract originator (the charitable organization) in order to transfer the money to the beneficiary.
- *TerminatetheCampaign()*: The contract originator or the Administrator can call this function. It sends the Ether back to donors and blocks any future interactions with this campaign.
- *VotetoTerminate()*: The function can be called by a donor who has donated to the campaign to vote for termination of the campaign.
- *GetSummary()*: This function gets called by the Dapp to display campaign details to the public.

Note that Metamask Ethereum wallet is required for all parties if they want to interact with the deployed contract, except, contribution or sending money to the campaign can be either by Metamask or other Ethereum wallets. Campaign smart contract rules are:

- Once the contract gets deployed, funds will be held in the smart contract in the Ethereum blockchain network till the campaign ends.
- When the campaign ends, only a charitable organization can finalize the contract to transfer the funds to the beneficiary address.
- A charitable organization cannot finalize the campaign without other parties' confirmation (Charity Evaluator and Beneficiary), should have passed campaign end

date, and the funds should reach the minimum amount of Ether.

- During the lifetime of the campaign contract, donors can get their Ether refunded at anytime they want.
- During the lifetime of the campaign donors can vote to terminate the campaign if they are not satisfied. If end date has already passed and 50% of the donors are not satisfied with the campaign the charitable organization (campaign originator) will not be able to finalize the contract, and they will be required to terminate the campaign.
- If the campaign end date has already passed, and the contract's conditions are not fulfilled, a charitable organization or Administrator can refund Donors money by calling the termination function.
- Once funds are sent successfully to the beneficiary, the campaign will not accept any new transaction, and no one can interact with it. However, all transactions and information will remain available and accessible to the public.
- Charitable organization (campaign originator) and Administrator can terminate the campaign at any time, when they do so, consequently, funds will be sent back to donors and the campaign will not accept any other transactions.

XI. MULTI-POINT MODEL

Fig. 7 illustrates how the Multi-point approach has been applied in the campaign smart contract; the campaign should go through seven processes, each of which is designed for a certain player.

For example when a charitable organization wants to finalize its campaign, the request through MetaMask wallet will be sent to the campaign smart contract in a transaction form to invoke the 'finalize function'. The campaign contract will first check if the sender ID is same as the contract originator ID to ensure that only the campaign originator (charitable organization) can finalize the campaign. It will then check if the campaign is validated by a charity evaluator organization, and verify that the beneficiary has confirmed the campaign and that 50% or more of donors are satisfied with the campaign. The campaign contract will then check if the campaign target is achieved and the campaign has reached its end date. Only when these conditions are met, will the transaction be confirmed and the campaign is finalized.

As shown in Fig. 6 the first step is to check the campaign status. If the campaign is finalized by its originator, users will no longer be able to interact with the finalized campaign. They can however continue to read the campaign details. The important point of this approach is to ensure that before the campaign is finalized or completed, it has to go through Multi-point campaign finalization process. This process would involve various parties in the fundraising process.

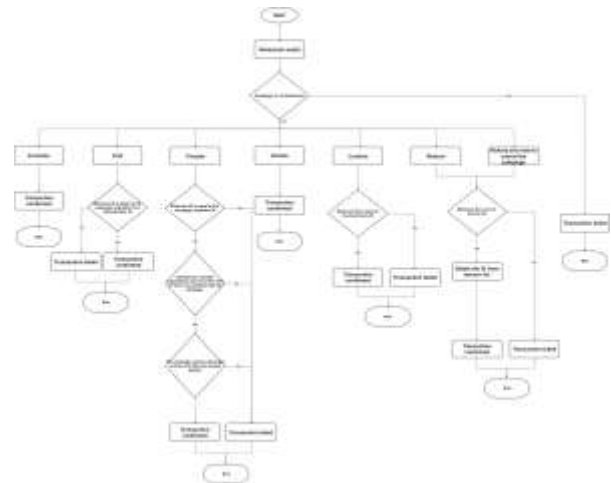


Fig. 7. Multi-point Model.

XII. SYSTEM INTERFACE

The system pages will be shown in this section, the system is a Dapp. The following points and figures illustrate the system pages, the Dapp interface consists of:

- Home page layout: Fig. 8 shows the home page of the Dapp. It is the first page that will be displayed for users - the first page that users will view in the website.

This page provides a short description about Blockchain and Smart Contract, as well as the latest two campaigns run by charitable organizations are displayed. From the navigation menu, user can go to the campaigns, about us, login and sign up pages.

- Campaigns page layout: as shown in Fig. 9 on this page users can check all campaigns that have been created by charitable organizations through the Campaign Factory smart contract. A list of campaigns will be displayed, consisting its campaign title, charitable name, short description about the campaign, and campaign status (to show if the campaign is completed).



Fig. 8. Home Page Layout.

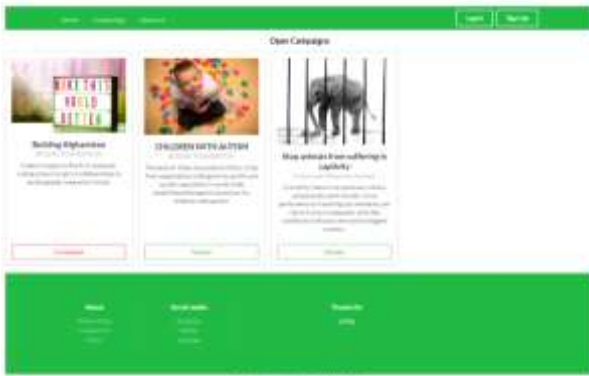


Fig. 9. Campaigns Page Layout.

- About us page: This page includes system directors' personal information such as name and email address. This page is also accessible to the website visitors to enable them to get in touch with the board members should the needs arise. Fig. 10 shows this page layout.



Fig. 10. About us Page Layout.

- Signup page layout: Fig. 11 illustrates this page layout. On this page the two types of users allowed to sign up are the charitable organization and charity evaluator organization. Users are required to fill fields that are divided into four main sections; first section is login information which includes user email and password. The second section includes organization name, mission and goals (for charitable organization) and standards (for charity evaluator organization), number of employees, address, and the registration number of the organization. The third section is the contact information such as phone number, email, and website. In the last section users are required to enter their Ethereum wallet address.
- Login page: as shown in Fig. 12 from this page users can access their accounts once their credentials are keyed in. The two fields that users are required to fill in are their email and password. If a user doesn't have an

account yet, he/she should go to the sign up page through a hyperlink provided underneath the fields form, or through the menu bar to create a new account.



Fig. 11. Signup Page Layout.



Fig. 12. Login Page Layout.

- Campaign details page: as Fig. 13 shows, this page displays all the details of a campaign; it is the main page of the Dapp since most of the interactions with the Campaign smart contract will take place on this page. Here, donors can donate to a campaign. The second section is used to show information about the originator of the campaign (charitable organization), which includes name and Ethereum address of the organization as well as a button to visit the organization's profile and a hyperlink to send users to the Etherscan to check its account. Details on the charity evaluator information are also displayed here. 'Waiting for validation from evaluator' message will be prompted if the campaign has yet to be validated by a charity evaluator. The beneficiary information is shown at the bottom of the page: as who is/are the beneficiary/ies, their physical address along with their Ethereum address and a hyperlink to visit their account in the Ethereum network. Underneath the beneficiary information section, unhappy donors could claim for a refund and vote to cancel the campaign. In addition, campaign information is shown under the campaign image - campaign title, short description about the campaign, and campaign status (it will be funded if the campaign is still running and completed if the campaign is completed and does not accept a new transaction). The minimum amount of funding needed for this the campaign is displayed with a progress bar

to show the percentage of the campaign balance, and donors rating (if a donor had requested for a refund of his Ether and voted to cancel the campaign hence affecting the rating). The same page also displays the campaign details such as start and end date, campaign target (the minimum amount), campaign balance, number of donors, number of unsatisfied donors, campaign manager name and email, campaign Ethereum address, and campaign goals.

Unlike the general users, the two types of logged in users (charitable organization and evaluator organization) can also see/access to the following buttons:

1) *Evaluator organization's button*: This button is for evaluator organizations to validate a campaign.

2) *Charitable organization's button*: Only charitable organizations can see this button to either finalize or terminate the campaign.

- Charitable organization's profile: After a user has verified his account and logged in, he needs to wait until his account gets approved by the admin. If approved, new campaign button will be appeared on the user's page as it is shown in Fig. 14 and he will then be able to update certain information such as address, number of employees, phone number, contact email, missions and goals, and email. The user will also be able to see his campaigns and access it to finalize, terminate, or just check on the campaign status.

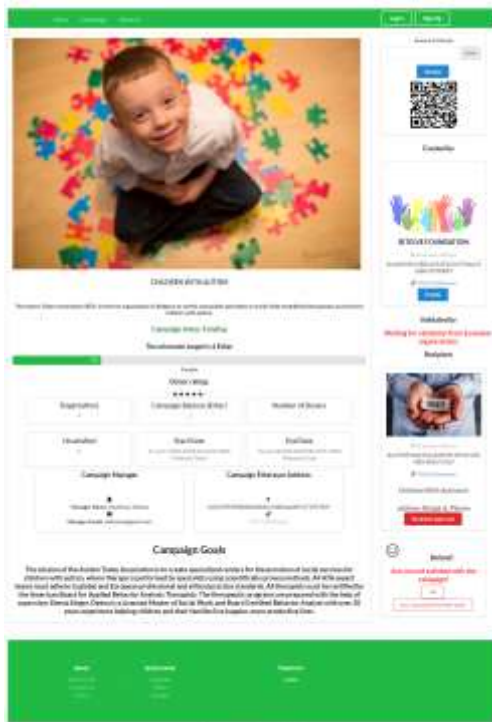


Fig. 13. Campaign Details Page Layout.



Fig. 14. Charitable Organization's Profile Layout.

As shown in Fig. 15, if the user wants to create a new campaign, a new page will be opened. This page is only accessible to the charitable organizations. To create a new campaign, the user will furnish some relevant information on the campaign, i.e. title, short description about the campaign, minimum target needed to fulfil campaign goals, beneficiary Ethereum address so that the funds will be automatically transferred to this account after the campaign is completed and finalized, campaign goals, beneficiary information, campaign manager information, and end date.



Fig. 15. Create New Campaign Page Layout.

The user must use his predefined Metamask address so that after clicking on the create button, his MetaMask wallet will pop up and display the transaction fee. Once the user clicks on the confirm button, only then will his new campaign be deployed in the Ethereum network.

- Charity Evaluator Organization: as Fig. 16 shows the charity evaluator organizations' profile is very similar to the charitable organization. After a user verifies his account and logged in, he needs to wait until he gets the admin's approval.



Fig. 16. Charity Evaluator's Profile Layout.

Similarly, when the user account gets approved, a campaign button will be appeared in his account and the user will be able to update certain information such as address, number of employees, phone number, contact email, standards, and email address.



Fig. 17. Charity Evaluator Campaigns Page Layout.

The user can also see those campaigns that have been validated by him as shown in Fig. 17. When the evaluator clicks on the campaigns buttons on the page, a list of campaigns will be displayed and the user can then access to information such as the campaign details and whether or not the campaigns are validated.

XIII. TEST SYSTEM'S SMART CONTRACT

The Dapp contracts have been tested locally; Rinkeby Ethereum test network is used during the test to obtain virtual Ethereum tokens for donation and expenses required for the test.

Infura provides a suitable entry point for the Rinkeby network. MetaMask wallet was used to store the Ethereum tokens and complete the corresponding transactions. At the same time, a Web3 instance completes the interaction between the website and the Ethereum network. Table I shows accounts and their balances that have been utilized for testing before the campaign is being deployed.

The campaign includes a description of the campaign, goals, information about the Originator and the Beneficiary, end date of the campaign to notify the public that the Originator will be able to finalize the campaign after that date, it also displays the campaign balance along with all transactions that have been made and the minimum target

amount needed to achieve campaign's goals (the minimum target for this campaign is 1 Ether).

Table II illustrates all interactions with System contracts; the first transaction made by Originator to create a new campaign. 0.00141 ETH of the gas limit was used to execute the create function in the campaign factory smart contract. The second transaction was made by the Donor A to send 0.25 Ether from his wallet to the testing campaign smart contract, the gas fee used to execute this transaction was 0.00012 Ether. Donor B also donated with Ether to the campaign at a different amount of 0.10 Ether. After that, the Originator tried to finalize the campaign, but the transaction failed because some requirements have not yet been fulfilled.

TABLE I. ETHEREUM ACCOUNTS THAT HAVE BEEN USED FOR TESTING THE SMART CONTRACT

Account	Address	Balance (ETH)
Originator	0x38e7be22eaFc465042f1b92c13D685342Bb034AC	1
Monitor	0xbB1E68cE914f95b4aB86d81581Bb25aB5C2B3402	1
Beneficiary	0x4a6F69e31BE1a5A3E4d7A794ba76a2Ba71f9DB93	0.0007
Donor A	0x21A1292D940090AB830eF88D0fb1891F0A220596	1
Donor B	0x0C69Ff3624c607Bc3Aca6BF8f2D5f93bb3B4bDD3	1
Donor C	0x2700556Ab9a1eb2D39c1D53C22b40E5982671457	1

TABLE II. ALL INTERACTIONS WITH THE CAMPAIGN SMART CONTRACT

Originator	Action	Message	Notes
Manager	Create a new campaign	Confirmed Transaction	-
Donor A	Donate (0.25 Ether)	Confirmed Transaction	Contract balance 0.25 Ether
Donor B	Donate (0.10 Ether)	Confirmed Transaction	Contract balance 35.0 Ether
Originator	Finalize the campaign	Failed Transaction	Failed transaction: because the contract has not been confirmed and verified by the beneficiary and the monitor.
Beneficiary	Approval	Confirmed Transaction	-
Originator	Approval	Confirmed Transaction	-
Manager	Finalize	Failed Transaction	Failed transaction: the end date has not passed yet, and the target balance has not reached.
Donor C	Donate (0.65 Ether)	Confirmed Transaction	Contract balance is 1 Ether
Originator	Finalize	Failed Transaction	Failed Transaction: because the end date has not passed yet.
Originator	Finalize	Confirmed Transaction	The contract finalized and the Ether sent to the beneficiary, no more interaction can be done.

The campaign includes a description of the campaign, goals, information about the Originator and the Beneficiary, end date of the campaign to notify the public that the Originator will be able to finalize the campaign after that date, it also displays the campaign balance along with all transactions that have been made and the minimum target amount needed to achieve campaign's goals (the minimum target for this campaign is 1 Ether).

Finally, the originator was able to finalize the contract and transfer the money to the Beneficiary when all the requirements and conditions have been fulfilled.

Table III presents the accounts' balance after the Testing Campaign has been finalized. All of the Testing Campaign's transactions with time and fees are recorded on Ethereum blockchain as shown in Fig. 18. The figure is taken from Etherscan website, a search engine that lets users look up, confirm and validate transactions on the Ethereum network.

TABLE III. ETHEREUM ACCOUNT AFTER THE CAMPAIGN SMART CONTRACT IS FINALIZED

Account	Address	Balance (ETH)
Originator	0x38e7be22eaFc465042f1b92c13D685342Bb034AC	0.998501
Monitor	0xbB1E68cE914f95b4aB86d81581Bb25aB5C2B3402	0.99997
Beneficiary	0x4a6F69e31BE1a5A3E4d7A794ba76a2Ba71f9DB93	1.000616
Donor A	0x21A1292D940090AB830eF88D0fb1891F0A220596	0.749872
Donor B	0x0C69Ff3624c607Bc3Aca6BF8f2D5f93bb3B4bDD3	0.89988
Donor C	0x2700556Ab9a1eb2D39c1D53C22b40E5982671457	0.349902

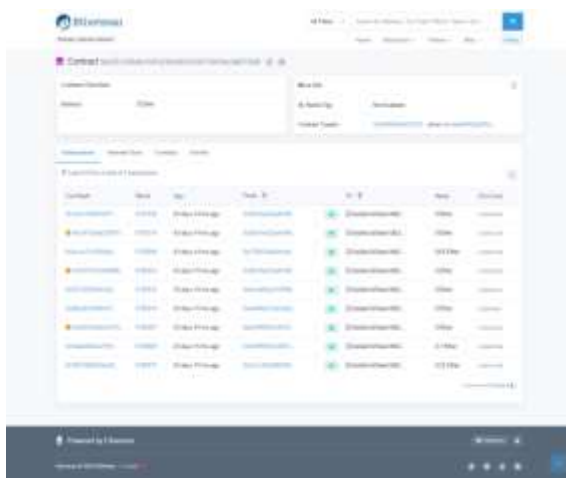


Fig. 18. Etherscan All Transaction Details.

XIV. DISCUSSION AND RESULTS

The study proposes the Multi-point approach to tackle transparency issue, in which the fundraising process should go through multi acceptance and specific conditions before it gets finalized. In other words, rather than having a campaign

managed and controlled by one party, this approach involves other parties between the charitable organization and the beneficiary and empower them to improve the transparency in fundraising. Thus, that could help in strengthening the relationship between the charitable organization and donors. Furthermore, it will create a transparent environment that could affect positively on the charitable organizations and increase the quantity and amount of donations to charities, as well as boost donors trust.

The biggest issue in the existing Dapps' campaigns is that they are fully controlled by one party which is the campaign's originator, so they can spend funds anytime and anywhere without any constraints. The proposed Dapp allows donors to play a significant role in the campaign. When the contract gets finalized by the originator the funds will be sent automatically to one direction (beneficiary address) that has already been defined in the campaign.

In addition, existing Dapps do not offer refund option nor do they ensure that the donated funds are sent to the beneficiary. On the other hand, the proposed Dapp ensures that donors can get their money back at any time during the lifetime of the campaign.

The Multipoint approach in fundraising is concerned about empowering donors and involving charity evaluator organizations in the fundraising, between a charitable organization and a beneficiary, in which, charity evaluator organizations are responsible for evaluating and monitoring charitable organizations' campaigns. With the presence of the charity organization evaluators who act as a party ensuring transparency and authenticity of a charity body, donors involved in the fundraising process are protected from fraudulent charities. Moreover, donors can evaluate campaigns and play a major role in the fundraising process. The Dapp interacts with two smart contracts: the Campaign Factory smart contract that uses for creating and tracking charities' campaigns, and the Campaign smart contract that creates by a charitable organization to receive and hold donors funds. Based on certain conditions such as the campaign status, charity evaluator's evaluation, and donor satisfaction, the smart contract determines whether or not the funds can be sent. Besides, all interactions (transactions) with the Campaign smart contract are accessible and traceable and they will be recorded in the Ethereum blockchain.

Finally, based on the above discussion the Dapps's smart contract has been tested and has proven that it can deal with the lack of transparency issue in charity, in consequence, rebuild trust and confidence in charity.

XV. CONCLUSION

Many people have a passion for contributing to society, and they want to donate generously to charitable organizations. However, the lack of transparency in charity caused a trust issue. Transparency is essential in fundraising to maintain public trust, and should be the top priority for charity organizations. Therefore, the research proposed a Multi-point

approach based on the usage of Blockchain technology to overcome the transparency issue in charity. First, it investigated the potentials of Blockchain can improve transparency, and then it analyzed the important features related to Blockchain that should exist in the charity's Blockchain platform. The significant features are public network, crypto-currency, smart contract, and consensus algorithm. Finally, the system was tested and proved that it can enhance transparency.

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Tracking Donations of Charitable Foundations Using Blockchain Technology

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Abstract: The paper looks at the possibilities of using blockchain technology for charitable purposes. To ensure data protection, fund integrity, and donation control, problems in this field necessitate the introduction of new storage tools and the transfer of information between donors, foundations, donation recipients, and other charitable actors. Via assured data protection and the ability to monitor the movement of funds and transactions, using the blockchain would increase the interest of potential donors in charitable organisations. The authors of this article examine needs and emerging blockchain-based charity networks in Russia and around the world. They provide an example of how distributed registry systems can be used to create a forum for making and monitoring charitable donations. The authors collaborated with local funds and non-profits during their research to validate the solution, gain a deeper understanding of ecosystem needs, and share their findings in the report.

Donors are suspicious of how their money is spent. Blockchain technology is currently being used in a variety of industries. You will use blockchain technology to make payments Transparency in the donation and fund-transfer process. It is necessary to create a single database for monitoring donations that will monitor all details about donations, transactions, and donors. The aim of this paper is to describe the implementation of a blockchain-based framework for monitoring donations. Centered on blockchain technology, the System provides open accounting of operations for donors, charitable foundations, and recipients.

should provide a clear donation route, allowing public users and donors to track and control where, when, and to whom charity funds were distributed.

Keyword— Charity Blockchain, Smart-contracts, Ethereum, Transparency

I. INTRODUCTION

According to National Research University's Higher School of Economics research, 57 percent of people donate. The proportion of Russian charitable donations in The GDP ratio is 0.34 percent. A donor has the right to request a report on funds spent; however, just 30% of donors follow the direction of their donations. However, the majority of donations are made informally. The money is given to the poor in person (alms, through family and friends, through work/study, or through a civil society initiative) and Formally, fundraising is not structured in nature, nor is it done on a continuous basis or with transparency. Donors seldom know how their funds were spent, even though they donated via a bank account, the Internet, or mobile contact (via SMS). This paper offers best practises for social intent architecture, platform design, and REST API implementation in blockchain projects. In Korea, a tradition of giving has emerged as a result of the rise of social consciousness. Transparency within a donation scheme, on the other hand, has long been a problem; for example, donors often want to know how their money is spent. However, a focus on transparency can cause donors and recipients to be concerned about their privacy. As a result, a donation mechanism should be built that ensures both transparency and privacy. Donors will not want their contributions to be revealed whether they collect them or give them to the donation system. If they used a donation system with a blockchain that included encryption, users would be able to build contracts and use the system with addresses that were not easily recognisable. However, in a blockchain system like this, the log can be examined to see if the same type of address performs the same action repeatedly. As a result, since the user's actions can be analysed, a privacy problem may arise. That is, the donation system will keep track of all contributions made or received, exposing personal information.

II. LITERATURE SURVEY

Paper Name- Blockchain-Based One-Off Address System to Guarantee Transparency and Privacy for a Sustainable Donation Environment

Author: Jaekyu Lee 1,[†], Aria Seo 2,[†], Yeichang Kim 3 and Junho Jeong 4,*

Description:

The problem of transparency in donation systems has long been a topic for discussion.

However, the emphasis on transparency raises privacy concerns for donors and recipients, with some people attempting to hide donations or the receipt of money. Therefore, a donation system that guarantees transparency and privacy is required to avoid negative side effects. In this study, we developed a system that protects personal information by using

a one-time account address system based on a blockchain while emphasizing transparency. The developed system could contribute to the creation of a sustainable and safe donation environment and culture.

2) **Paper Name-** Developing a Reliable Service System of Charity Donation during the Covid-19 Outbreak

Author-Hanyang WU1, and Xianchen Zhu2

Description:- Drawing upon the functional characteristics of blockchain technology, this paper envisages the feasibility and reliability of developing a charity donation service system loaded onto blockchain in response to the complex service demands encountered by charity operators due to the Covid-19 epidemic. With blockchain technology's support as the underlying data book, this paper focuses on the practical issues of charity donation fund and material allocation, as well as information release and sharing, charity donation organization, and organization self-management. The paper thereby discusses the key technologies in terms of overall structure design, specific service sector, and functional design of the donation service system and further summarizes the operational mechanism of the system as combined with the needs of help-seeking, receiving, and management users. It is argued that all the above proposals have the potential to alleviate the trust crisis of charity services in China in view of low transparency. The paper expects to provide a useful reference for charity business innovation propelled by blockchain technology.

3) **Paper Name-** Platform for Tracking Donations of Charitable Foundations based on Blockchain Technology

Author:-Azamat Dzhonov

Description- Donors have distrust about how donated money is spent. Currently, blockchain technology is being implemented in different sectors. Blockchain technology allows you to make the process of donations and transactions of funds transparent. Single platform for tracking donations that will track all information about donations, transactions and donors need to be developed. This paper considers description of implementation of the platform for tracking donations based on blockchain technology. The System offers transparent accounting of operations donors, charitable foundations and recipients based on blockchain technology, charitable platform should provide transparent donation route, enable public users and donors to track and monitor where, when and to whom went resources of charity funds.

4) **Paper Name:-**Proposed Solution for Trackable Donations using Blockchain

Author- N. Sai Sirisha, Tarasha Agarwal

Description- The lack of transparency has made people lose trust in charities, making social funding stagnant. The donor is unaware of the legitimate utilization of his funds. Corruption adds to the distrust of the donor. This paper proposes a system called Charity-Chain that is a decentralised network built on the Ethereum blockchain. It helps social organisations to run projects transparently, using smart contract-based incentives to ensure their impact is independently verified and accessible to everyone. This makes it much easier for funders (philanthropic organisations, impact investors, small donors) to monitor their transactions and hence restore their trust in giving to such social organizations.

III. PROPOSED METHODOLOGY:

A. PROPOSED-SYSTEMARCHITECTURE

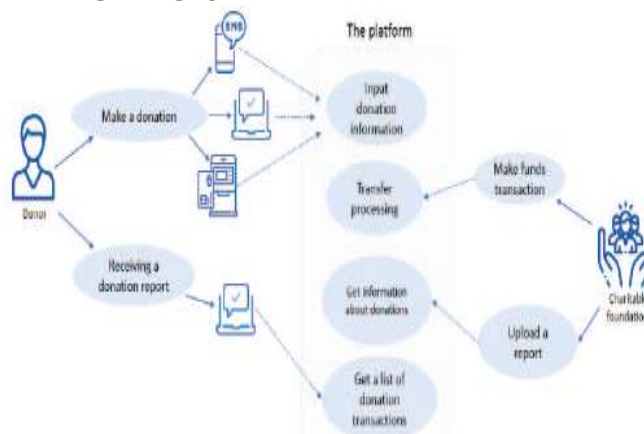


Fig:-system architecture

The architecture of a decentralized blockchain-based application. Examples of such applications can be Bitcoin. In this case, the data in the network is distributed, and all business logic is implemented in the blockchain. However, a large amount of data is created in the donation field, including information about donations, transactions, and donors. At the moment, data writing to blockchain is not possible. needs disc space on all network nodes, and recording takes a long time. This causes issues in our project, which has a broad data flow. The vast majority of blockchain-based projects aren't fully decentralised. They are usually tightly integrated with a client/server architecture (hybrid). Outside of the blockchain, all minor application data is stored in a centralised storage system. The master data is kept in a database. The term "blockchain" refers to a Smart contracts are used to create all entries in the decentralised storage. REST requests are used to send and receive data from the blockchain and centralised storage.

V. CONCLUSION AND FUTURE WORK

To address the issues with current sponsorship schemes, we create a donation system that can improve accountability while maintaining privacy. The useP2P-mixing-technique-based blockchain is d to build the method. As a result, other donors and receivers are unable to identify donors and receivers. When submitting donations, the process of converting an account address. The algorithm for generating the detailed structure of the system and updating the address is described in this section. The donor is a machine owner with a private key and a public key who makes donations. The donor account is one that the donor creates with the help of the public and private keys. The donor is the owner of the donor address (Donor Eth Addr) provided by the account. Smart contracts can be created and accessed using addresses. The recipient is a member of the system who accepts donations and has both a private and public key. The receiver account is formed with the receiver's private and public keys, and it owns the receiver's address (Receiver Eth Addr).

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